

Chapter 6

Superconductivity

6.1 Introduction

In solid state physics, many of the phenomena that we are familiar with occur because of the interparticle interactions among the charge carriers, thereby resulting in an ordered state. The effects of such interactions are best perceptible at low temperature. With increase in temperature, the thermal motion of the carriers gains prominence, and outdoes the ordering process. Superconductivity denotes one such ordered phase of matter which stabilizes at low temperature, and vanishes when the temperature is increased beyond a certain critical point.

In the year 1908, H Kamerlingh-Onnes [1] at the low temperature lab in Leiden, Netherlands successfully liquified helium (He). At normal atmospheric pressure, the boiling point of He was found to be 4.2 K, thereby making exploration of material properties feasible at low temperatures. Studying electronic conductivity (or resistivity) of metals seemed like a normal choice, as metals are primarily characterized by their electrical resistance. Measurements done at low temperatures may yield the following possibilities:

- (i) the resistance vanishes gradually with decreasing temperature;
- (ii) it may result in a small but finite value at very low temperature;
- (iii) it may have a minimum at low temperature, and finally show an upturn before diverging at very low temperatures.

In particular, the last possibility receives support on physical grounds in the sense that at sufficiently low temperature, the carriers are likely to be bound to their respective atoms. Thus their ability to move around and contribute to the conductivity vanishes.

K Onnes realized the importance of studying conductance characteristics of high purity metals at low temperatures. Initially he started with gold (Au) and platinum (Pt), mainly because they are noble metals and available in pure form. However, he shifted his attention towards mercury (Hg) which can be obtained in a highly pure form via multiple distillations. Precisely at the boiling point of liquid He, he found that the resistance of an ultra-clean Hg sample sharply vanishes, and at further lower temperatures, the resistance becomes immeasurably small (see figure 6.1). The temperature at which the resistance disappears is called the transition temperature, T_c . Below that a new state of matter emerges, which either completely expels the magnetic field or, in some cases, traps magnetic flux having values that are in integer multiples of the flux quantum, Φ_0 ($\Phi_0 = 2.07 \times 10^{-15}$ Wb). This was indeed a surprise, since laws of electromagnetic induction predict that ideal conductors retain, and do not expel the magnetic field that is trapped inside.

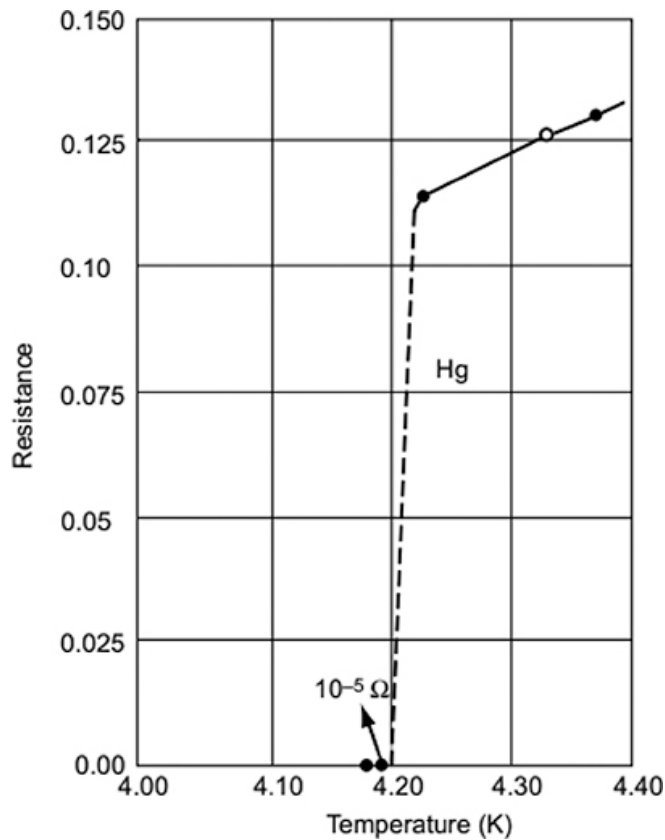


Figure 6.1. The fall of the electrical resistance of Hg (in Ω) as a function of temperature is shown. Near 4.2 K, there is a sharp fall of resistance, below which the resistance becomes almost zero (after K Onnes' original experiment [1]).

A complete theoretical understanding of the microscopic phenomena had to wait about half a century till the theoretical derivation by Bardeen, Cooper and Schrieffer (BCS) in 1957 came into existence, for which they were awarded the Nobel prize in 1972. They realized at the transition point, the electrons pairwise condense into a new phase which is a coherent matter wave with well-defined phase relationship among the pairs. These electrons form pairs mediated via quantized lattice vibrations, namely the phonons.

For nearly eight decades after the experimental discovery of superconductivity in Hg in 1908, it remained a low temperature phenomena till Bednorz and Müller discovered the onset of superconductivity in the copper oxide planes at larger temperatures. As the years passed, discoveries of a large number of copper oxide superconductors were made with larger and larger critical temperatures, which was thought to have far reaching consequences in terms of their industrial applications. There was moderate success on that front, however, there were many surprises, which arose as the experimental data started coming in on the physical properties of these superconductors. Quite likely, the normal state¹ of these copper oxide materials is very different than those for the conventional superconductors that were

completely understood within the BCS theory. Since the beginning point of building up a theory, which in this case is the normal state, is missing, or at least, is unlike the conventional metallic state, a complete understanding of these superconductors remained elusive.

6.1.1 Historical developments

Empirical rules have been suggested from time to time to enable identification of possible new superconducting materials. Rules proposed by Matthias [2] in the 1960s are especially noteworthy. One of them is based on the number of valence electrons. It is found that the materials with on an average 5 to 7 electrons per atom show relatively high T_c 's. For example, RuMo with an average of 7 electrons shows $T_c = 10.6$ K, whereas, ruthenium (Ru) with an average of 8 electrons shows $T_c = 0.5$ K, and molybdenum (Mo) with an average of 6 electrons shows $T_c = 1$ K. Later on, a structural dependence was seen on T_c . A-15 structure or the β -tungsten shows high T_c . V_3Si and Nb_3Ge are examples of this structure and they show $T_c = 17.1$ K and 23.2 K, respectively. For quite a few years, this 23.2 K barrier was never crossed. Among the organic materials, the intercalated compounds of graphite (e.g. C_8K) become superconducting below 1 K, while K_3C_{60} has $T_c \approx 18$ K. Generally alkali salts of C_{60} have been found to be superconducting with T_c ranging from 18 to 35 K.

If we refer to the periodic table and consider the top row of the middle section of the periodic table, that is, group IVB (Cr, Mn) till group VIII (Fe, Co, Ni), they have strong magnetic character. While along the same row, Ge, As, Se or some of the neighbouring elements show superconductivity. Since both the phenomena have electronic origin, and their physical properties are quite distinct, it forces us to wonder what could be the genesis of this difference. It probably lies in the extent of the wavefunction of the electrons: for magnets, the wavefunction has small spatial extent, that is spread over a smaller number of atoms, while for superconductors, the spread is large, thereby promoting pairing phenomena between the carriers.

6.1.2 Physical properties

Before we proceed to discuss the physical properties of superconductors, we wish to make the plans clear for our upcoming discussion. We shall mainly mention salient features of the superconductors that make them worthy of study. However, most of the ideas and concepts are introduced with minimal amount of derivation. For a thorough understanding of some of the properties, one has to wait for the BCS theory, a topic that we discuss somewhat elaborately afterwards.

Let us rewind the ongoing discussion for more details. The resistance of a metal drops to zero very sharply, within a temperature window of $\Delta T \sim 10^{-5}$ K below a certain critical temperature, T_c . The superconducting state is characterized by zero electrical resistance. Besides, there is no change in crystal structure as verified by x-ray diffraction above and below the transition temperature. Thus to list out the properties, the state is characterized by:

- (i) infinite electrical conductivity, that is, $\sigma \rightarrow \infty$;
- (ii) finite current density, $j = \text{finite}$;

- (iii) the specific heat as a function of temperature shows a jump at $T = T_c$, and thus the transition to the superconducting state involves latent heat, and hence the transition from the superconducting state to the normal state is a second-order phase transition;
- (iv) vanishing values of the electric field, $E \rightarrow 0$ inside the superconducting sample;
- (v) constant magnetic field, $B \rightarrow \text{constant}$ inside a superconductor.

Clearly the last two statements defy the laws of classical electrodynamics. This can be understood in the following way. Using Ohm's law one can write, $\mathbf{j} = \sigma \mathbf{E}$ for j having a finite value and $\sigma \rightarrow \infty$, E has to be zero. Hence, using Maxwell's equation,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \text{or,} \quad \mathbf{B} = \text{constant.}$$

6.1.3 Meissner effect

The complete and sudden vanishing of the electrical resistance below a certain temperature raised a lot of questions. Some physical phenomena that were operative to render electrical resistance above T_c , were suddenly becoming ineffective. Meissner and Ocshenfeld (1933) meanwhile discovered that a superconductor completely excludes an external magnetic field. They measured the magnetic field distributed outside the superconducting materials (such as lead (Pb) or tin (Sn)) which are cooled below their respective transition temperatures while the magnetic field is switched on. The corresponding results could not be explained by superconductors being just resistance-less metals.

Above a certain critical magnetic field (of the order of a few Oersted), there is no expulsion of the magnetic flux, where superconductivity disappears and the material reverts back to its normal resistive state and the magnetic field fully penetrates it. This also endorses a close link between magnetism and superconductivity.

The exclusion of the magnetic field from a superconductor takes place regardless whether the material becomes superconducting before or after the external field is applied. At equilibrium, the external field is cancelled in its interior by screening fields produced by the skin current.

The total exclusion of the magnetic field from inside the superconductors assigns a property known as *perfect diamagnetism*. It can be understood as follows,

$$B = \mu_0(H_{\text{ext}} + M).$$

Since $B = 0$ inside a superconductor, the induced magnetization M cancels the external field, H_{ext} . Thus, the magnetic susceptibility, χ becomes,

$$\chi = \frac{M}{H_{\text{ext}}} = -1.$$

Thus, no known material is more diamagnetic than a superconductor. Typical diamagnetic susceptibility of metals is about 10^{-5} – 10^{-6} . Thus an usual diamagnetic material only expels a small fraction (about 10^{-5}) of the external field, while a superconductor does it completely.

However, even if we claim that the magnetic field is totally expelled from a superconductor, the fact is that it enters only up to a certain distance, called penetration depth.

6.1.4 Perfect conductors and superconductors

In order to differentiate a perfect conductor (which may show a gradual vanishing of the resistance as $T \rightarrow 0$), and a superconductor, we resort to field-cooled (FC) and zero-field-cooled (ZFC) techniques that are schematically shown in figures 6.2 and 6.3. If we consider cooling perfect conductors and superconductors in the absence of a magnetic field (left panels of figures 6.2 and 6.3), they behave identically when an external magnetic field is applied. That is, they expel the field by developing surface current which prohibits the field lines from penetrating beyond a certain distance (discussed below). However, in the presence of an external field, the behaviour of a conductor and that of a superconductor are quite distinct. When a conductor is cooled to a perfect conducting state in the presence of an external field, the flux lines get trapped inside, while for a superconductor the flux lines are expelled. Upon withdrawal of the field (panel (g) in figures 6.2 and 6.3), a superconductor is left with no memory, where for a perfect conductor, the flux lines form a loop, and thus it retains the field in its memory.

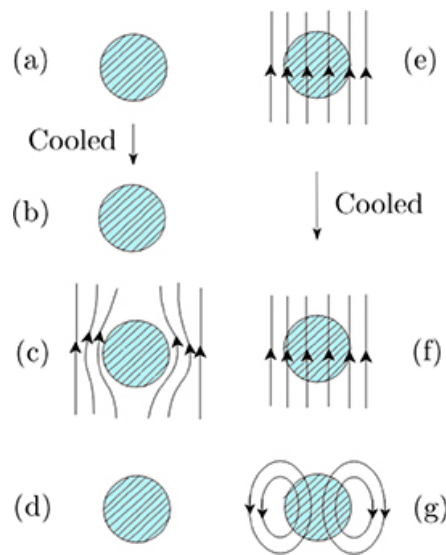


Figure 6.2. The ZFC (left panel) and FC (right panel) cases for a perfect conductor are shown schematically.

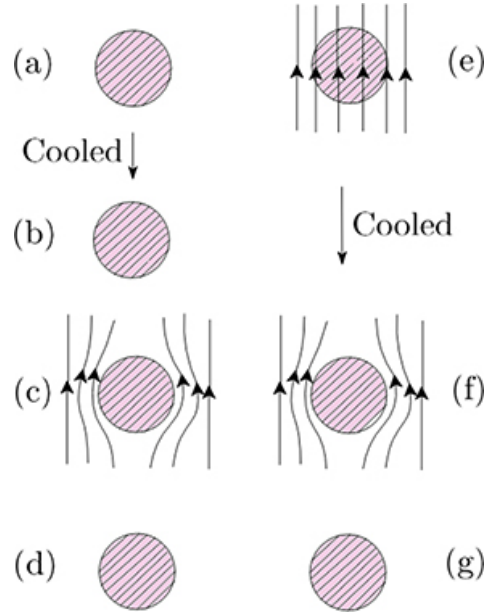


Figure 6.3. The ZFC (left panel) and FC (right panel) cases for a superconductor are shown schematically.

6.1.5 Electrodynamics of superconductors: London theory

Before the discovery of Meissner effect in superconductors, Becker *et al* [3] analysed the electrodynamic behaviour of perfect conductors using a simple free-electron model. According to it, the electrons accelerate under the application of an electric field, $\mathbf{E}$. The argument for resistance-less motion is that whenever the electrons faced a resistance, the applied electric field would accelerate them steadily according to Newton's laws,

$$m^* \frac{d\mathbf{v}}{dt} = e^* \mathbf{E} \quad (6.1)$$

where m^*, e^* are the effective mass and the net charge, respectively. It was found later that $e^* = 2e$ and $m^* = 2m$, which were subsequently called superelectrons or Cooper pairs. If there are n_s numbers of superelectrons moving with a mean local velocity $\mathbf{v}_s$, then the supercurrent density is given by,

$$\mathbf{J}_s = n_s e^* \mathbf{v}_s = -n_s |e^*| \mathbf{v}_s \quad (6.2)$$

The minus sign signifies that the supercurrent moves in an opposite direction to that of the superelectrons. Substituting equation (6.2) in equation (6.1) one gets,

$$\mu_0 \lambda_L^2 \frac{\partial \mathbf{J}_s}{\partial t} = \mathbf{E}$$

where, $\lambda_L = \sqrt{\frac{m^*}{\mu_0 n_s (e^*)^2}}$. λ_L has the dimension of length. Equation (6.3) is called the first London equation. Taking the curl of both sides of equation (6.3),

$$\mu_0 \lambda_L^2 \frac{\partial}{\partial t} (\nabla \times \mathbf{J}_s) = \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (6.4)$$

Using the following Maxwell's equation,

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}$$

we have,

$$\begin{aligned} \lambda_L^2 \frac{\partial}{\partial t} (\nabla \times (\nabla \times \mathbf{B})) &= -\frac{\partial \mathbf{B}}{\partial t} \\ \lambda_L^2 \nabla^2 \left(\frac{\partial \mathbf{B}}{\partial t} \right) &= \frac{\partial \mathbf{B}}{\partial t}. \end{aligned} \quad (6.5)$$

This implies that the rate of the $\mathbf{B}$ field (or the $\mathbf{B}$ field itself) will fall off exponentially inside a superconductor to a trapped field B_0 . This clearly contradicts the Meissner effect, that is, the magnetic field inside the specimen is zero, irrespective of the initial condition.

Thus the brothers F London and H London [4] suggested that since the macroscopic theory of a perfect conductor makes a correct prediction about superconductor for the special case $B_0 = 0$, thus it might be reasonable to assume that the magnetic behaviour of a superconductor may be correctly described according to Meissner effect not only to $\frac{d\mathbf{B}}{dt}$ but also to $\mathbf{B}$ itself, that is,

$$\begin{aligned} \lambda_L^2 \nabla^2 \mathbf{B} &= \mathbf{B} \quad [\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s] \\ \text{which implies, } \mathbf{B} &= -\mu_0 \lambda_L^2 \nabla^2 \mathbf{J}_s. \end{aligned} \quad (6.6)$$

By introducing a magnetic vector potential defined by $\nabla \times \mathbf{A} = \mathbf{B}$ and choosing a proper gauge, such as a transverse gauge, namely, $\nabla \cdot \mathbf{A} = 0$, the current density can be written as,

$$\mathbf{J}_s = -\frac{1}{\mu_0 \lambda_L^2} \mathbf{A} \quad [\mathbf{B} = \nabla \times \mathbf{A}]. \quad (6.7)$$

This is called the second London equation, and is reminiscent of Ohm's law. This is of course true for simply connected superconductors. The gauge condition demands

$$\nabla \cdot \mathbf{A} = 0 \Rightarrow \mathbf{A} \cdot \hat{n} = 0$$

on the boundary surface of the superconductor, $\hat{n}$ being the unit drawn normal.

Now, using Maxwell's equation,

$$\begin{aligned}
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \left(\mathbf{E} - \mu_0 \lambda_L^2 \frac{\partial \mathbf{J}_s}{\partial t} \right) &= 0 \\
\text{or, } \mathbf{E} - \mu_0 \lambda_L^2 \frac{\partial \mathbf{J}_s}{\partial t} &= \nabla \phi
\end{aligned} \tag{6.8}$$

ϕ is a scalar. One important comment is due here. There is no component of $\nabla \phi$ in the direction of $\mathbf{J}_s$, that is,

$$\nabla \phi \cdot \mathbf{J}_s = 0 \tag{6.9}$$

as $\mathbf{J}_s \cdot \mathbf{E} = 0$ from first London equation (6.3). This is correct as otherwise it would mean energy dissipation in a constant magnetic field which we know to be incorrect as it violates conservation of energy.

6.1.6 Penetration depth

Let us prepare a setup to solve equation (6.5) for a superconductor. Refer to figure 6.4 where a semi-infinite superconducting sample is placed along the y -axis. We are interested in studying the variation of the magnetic field only in the x -direction along which it has a finite width (axis shown in figure 6.4). In figure 6.4 the variation is expected to be along the x -direction, we can convert it into a one-dimensional equation, that is, by assuming,

$$\frac{\partial B}{\partial y} = \frac{\partial B}{\partial z} = 0$$

which yields,

$$\frac{dB_x}{dt} = \alpha \frac{\partial^2}{\partial x^2} \left(\frac{dB_x}{dt} \right).$$

Integrating over time yields,

$$\begin{aligned}
B_x &= \alpha \frac{\partial^2}{\partial x^2} B_x \\
B(x) &= B_{\text{ext}} e^{-x/\lambda_L}
\end{aligned}$$

where an exponentially growing solution, even though mathematically admissible, is dropped, and

$$\alpha = \lambda_L^2$$

λ_L is of the order of $500 \text{ \AA}$ for commonly known superconductors. This is how the magnetic field varies inside a superconducting sample. At $x = 0$, the value of the magnetic field just outside the surface is B_{ext} , which decays exponentially inside the sample.

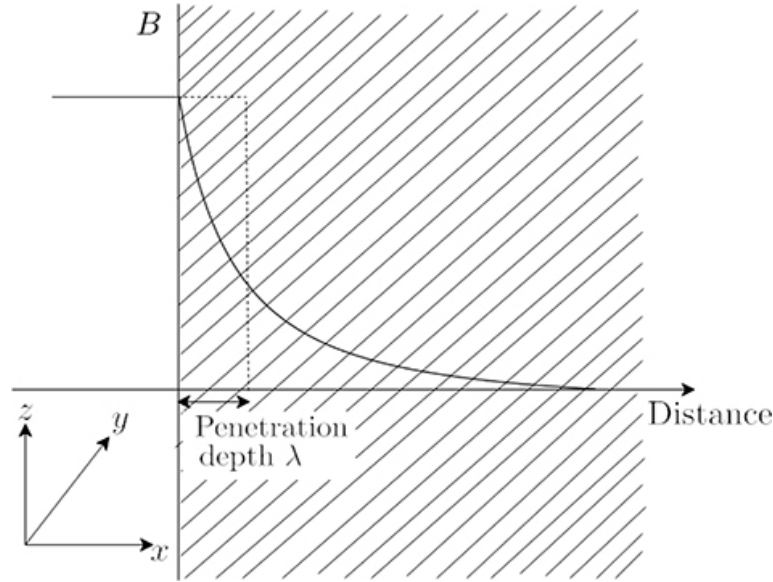


Figure 6.4. Schematic demonstration of the behaviour of an external magnetic field is presented. The field strength falls by a factor $1/e$ inside the superconductor, which is defined as the penetration depth.

6.1.7 Flux quantization

In the mid-1930s [4] it was realized that superconductivity is inherently a quantum phenomenon that manifests at macroscopic scales. The description thus requires a wavefunction which can be written as,

$$\psi(\mathbf{r}, t) = |\psi| e^{i\theta} \quad (6.10)$$

where θ is a real scalar function representing the phase of the wavefunction and $|\psi|^2 = n_s$, where n_s is the density of superelectrons. The canonical momentum for a particle of charge e^* and mass m^* in a magnetic field is given by,

$$\mathbf{p} = m^* \mathbf{v}_s + e^* \mathbf{A}. \quad (6.11)$$

Putting $\mathbf{p} = -i\hbar \nabla$ and writing down the Schrödinger equation,

$$\hbar \nabla \theta \psi = (m^* \mathbf{v}_s + e^* \mathbf{A}) \psi \quad (6.12)$$

yields,

$$\frac{\hbar}{e^*} \nabla \theta = \mu_0 \lambda_L^2 \mathbf{j}_s + \mathbf{A}$$

where $\lambda_L = \sqrt{\frac{m^*}{\mu_0 n_s e^{*2}}}$ is the London penetration depth (described later), and $\mathbf{j}_s$ is the supercurrent density defined via, $\mathbf{j}_s = n_s e^* \mathbf{v}_s$. Since $\mathbf{A}$ enjoys gauge freedom, the (local) phase of the wavefunction can be changed in the following manner, so as to keep the velocity unchanged.

$$\theta \rightarrow \theta' = \theta + \frac{e^*}{\hbar} \mathbf{A} \cdot \mathbf{r}$$

It may be noted that equation (6.13) is consistent with Maxwell's equation, which can be checked by taking a curl of the equation, and noting that $\mathbf{B} = \nabla \times \mathbf{A}$. Now taking a derivative of equation (6.13) with respect to time, one gets,

$$\frac{\hbar}{e^*} \nabla \left(\frac{\partial \theta}{\partial t} \right) = \mu_0 \lambda_L^2 \frac{\partial \mathbf{j}_s}{\partial t} + \frac{\partial \mathbf{A}}{\partial t}. \quad (6.14)$$

Using the first of the London equations, namely,

$$\frac{\partial \mathbf{j}_s}{\partial t} = \frac{\mathbf{E}}{\mu_0 \lambda_L^2}$$

one arrives at

$$\mu_0 \lambda_L^2 \frac{\partial \mathbf{j}_s}{\partial t} = \mathbf{E}. \quad (6.15)$$

Hence, with

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}$$

one gets,

$$\frac{\hbar}{e^*} \frac{\partial \theta}{\partial t} = -\phi. \quad (6.16)$$

Thus the time variation of the local phase denotes a scalar potential ϕ .

There is a feature of the superconductors that deserves special attention, which is quantization of the magnetic flux. Consider a multiply connected superconductor, and consider a closed loop C lying entirely in the superconducting regime, and encircling the hole (marked in figure 6.5). Thus any open surface bounded by C contains regions that are partially superconducting, and partially in a non-superconducting (normal) region. Applying Stokes theorem on equation (6.13) one obtains,

$$\frac{\hbar}{e^*} \oint_C \nabla \theta \cdot d\mathbf{l} = \Phi_s + \oint_C \mu_0 \lambda_L^2 \mathbf{j}_s \cdot d\mathbf{l}$$

Φ_s is the magnetic flux defined by, (6.17)

$$\Phi_s = \oint \mathbf{B} \cdot d\mathbf{s}. \quad (6.18)$$

The requirement that ψ is a single valued function at each point, that is,

$$\phi e^{i\theta(\mathbf{r},t)} = \phi e^{i\{\theta(\mathbf{r},t)+2n\pi\}}$$

indicates that the LHS of equation (6.17) has the form,

$$\frac{\hbar}{e^*} \int_c \nabla \theta \cdot d\mathbf{l} = n\Phi_0 \quad n = 0, \pm 1, \pm 2, \dots \quad (6.19)$$

where Φ_0 is called the flux quantum

$$\Phi_0 = \frac{2\pi\hbar}{|e^*|} = 2.07 \times 10^{-15} \text{ Wb} \quad (6.20)$$

$n = 0$ corresponds to simply connected superconductor.

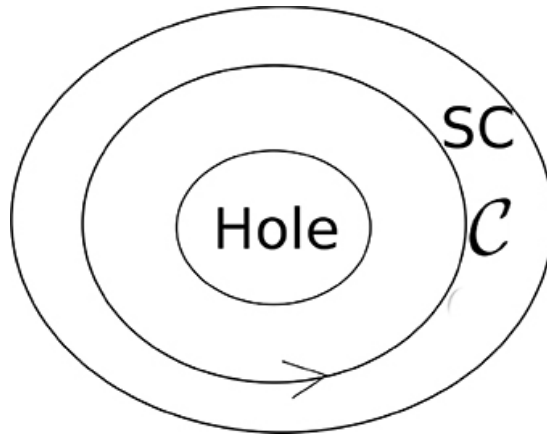


Figure 6.5. A simply connected superconductor is shown with a ‘hole’ within. C denotes a contour of integration in equation (6.13).

Hence integrating over a closed loop C should be the same after one complete rotation from where one has started, except for a phase of 2π , or a multiple of 2π . A flux quantum has this value for $e^* = 2e$. So the supercurrent is carried by a pair of electrons. In fact,

$$\oint \nabla \theta \cdot d\mathbf{l} = 2\pi n = -\frac{2e}{\hbar} \Phi.$$

Thus the magnetic flux penetrating through a superconductor can only appear in multiples of the superconducting flux quantum Φ_0 , which is often referred to as fluxoid (or fluxon). It may be noted that the value of the fluxoid is extremely small. Experimentally, the flux quantization was measured by Deaver and Fairbank in 1961 [5]. Importantly, the involvement of Planck's constant ($\hbar$) is a signature of quantum effects existing at macroscopic scales.

In the right-hand side of equation (6.17) we can introduce a fluxoid Φ_L ,

$$\Phi_L = \int_s \mathbf{B} \cdot d\mathbf{s} + \oint_c \mu_0 \lambda_L^2 \mathbf{j}_s \cdot d\mathbf{l} \quad (6.21)$$

Thus within approximation of the London theory,

$$\begin{aligned} \frac{\partial \Phi_L}{\partial t} &= \frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s} + \oint_c \mu_0 \mathbf{E} \cdot d\mathbf{l} \\ &= \int_s \left(\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} \right) \cdot d\mathbf{s} \\ &= 0 \end{aligned} \quad (6.22)$$

where the last step is obtained using Maxwell's equation. Thus the fluxoid Φ_L is constant with time. It also does not depend upon the exact shape of the contour C , as long as it encompasses the hole, but only once. If there is no hole, then the fluxoid is zero.

6.1.8 Non-local electrodynamics

The current density equation in electrodynamics, namely,

$$\mathbf{J}_s = n_s e \mathbf{v}_s$$

is a local equation in the sense that it relates the current density at a point $\mathbf{r}$ to the velocity of the charge carriers at that point. Thus it clearly ignores the spatial structure of the electric field. An improvement to this can be done by considering the current density at $\mathbf{r}$ due to the variation of the electric field within a shell around the point $\mathbf{r}$, with a finite radius. On similar grounds, Pippard argued that the current density at a point $\mathbf{r}$ should depend on $\mathbf{E}(\mathbf{r}')$. Further he conjectured that the wavefunction of the electrons must have a characteristic dimension, namely, ξ which is called the coherence length. A simple uncertainty calculation may be sufficient to yield an estimate of the value of ξ in typical superconductors. We know that electrons within an energy range $k_B T_c$ play a dominant role in the pairing phenomena. The momenta of these electrons have an uncertainty given by,

$$\Delta p \simeq \frac{\Delta E}{v_F} \simeq \frac{k_B T_c}{v_F}.$$

Thus, their position uncertainty, using Heisenberg's uncertainty principle is,

$$\Delta x \simeq \frac{\hbar}{\Delta p} \simeq \frac{\hbar \nu_F}{k_B T_c}.$$

This yields,

$$\Delta x \simeq \xi = a \frac{\hbar \nu_F}{k_B T_c}$$

a is usually of the order of unity. In particular, $a = 0.8$ in BCS theory [6].

Hence, Pippard suggested that the simple equation for the current density written above should be modified as,

$$J_s(\mathbf{r}) = -(\text{const}) \int \frac{\mathbf{R}[\mathbf{R} \cdot \mathbf{A}(\mathbf{r}')] }{R^4} e^{-R/\xi} d\mathbf{r}' \quad (6.23)$$

where, $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, with $\mathbf{r}$ being the source point and $\mathbf{r}'$ being the field point, or where the observation is made.

6.2 Magnetic phase diagram of superconductors

When a superconductor is used to form a circuit with a battery, and a steady state is established, all the currents passing through the superconductor are supercurrents. Normal currents due to motion of charged particles contribute zero, since no voltage difference can be sustained in a homogeneous superconductor (because otherwise a current will induce a B which is zero). Experiments show that all supercurrents flow near the type-I superconductor's surface within a thin layer characterized by the penetration depth. These surface supercurrents run so that the B field is cancelled in the interior of a conductor.

Thus, let us review the magnetic phase diagram of superconductors. It is clear by now that the superconducting phase is a stable phase in a certain range of magnetic fields and temperatures. For higher fields and temperatures, the normal metallic state becomes more stable. The critical magnetic field (H_c) beyond which the material cannot be superconducting even if it is cooled below T_c (see figure 6.6). The value of H_c depends on temperature, since it is easier to quench a superconductor at higher temperatures, than at lower temperatures. An empirical relation for the same can be written as,

$$H_c = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right] \quad \text{where, } H_0 = H_{\text{ext}}(T = 0). \quad (6.24)$$

The corresponding M versus H phase diagrams are shown in figure 6.7. There is a sharp fall of the (negative) magnetization ($-M$) as a function of the external field, and a sharp rise of the magnetic induction denotes a phase transition from a superconducting to a normal phase.

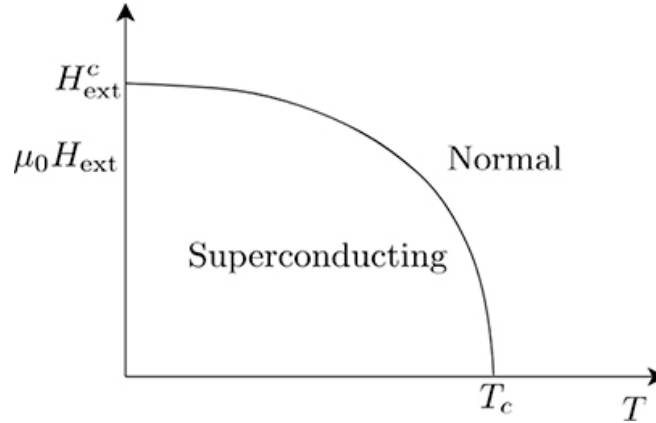


Figure 6.6. The dependence of the critical field, H_C as a function of temperature T is shown.

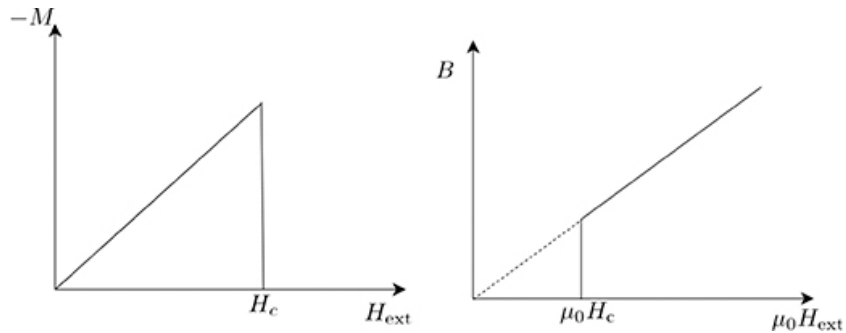


Figure 6.7. Magnetic phase diagrams for type-I superconductors are shown. There is a direct transition from the superconductor to the normal phase in type-I superconductors.

In type-II superconductors, the phase diagram is more complex. When small magnetic fields are applied, surface currents develop that screen the magnetic field from penetrating homogeneously into the sample. This is termed as the Meissner state, which persists till a certain critical value, called the first critical magnetic field, H_{c1} . Beyond this value, the system enters into a mixed state, where the magnetic flux lines, in the form of vortices, penetrate the superconducting sample. Thus there is a partial screening of the magnetic field.

Beyond H_{c1} , more and more vortices appear insides the superconductor, and eventually when the external field reaches a second critical value, namely, H_{c2} the material becomes a normal metal (figure 6.8). The mixed phase is called the Abrikosov phase, where it is energetically favourable for the vortices to self-organize into a hexagonal array. The scanning

tunnelling microscopy (STM) measurements² confirm the existence of such a vortex array, which minimize energy by arranging into hexagonal structure in which the vortices line up. These structures are called Abrikosov lattices. The magnetic susceptibility of a superconductor is -1 below H_c and has a value 10^{-5} above H_c . This is shown in the following figures.

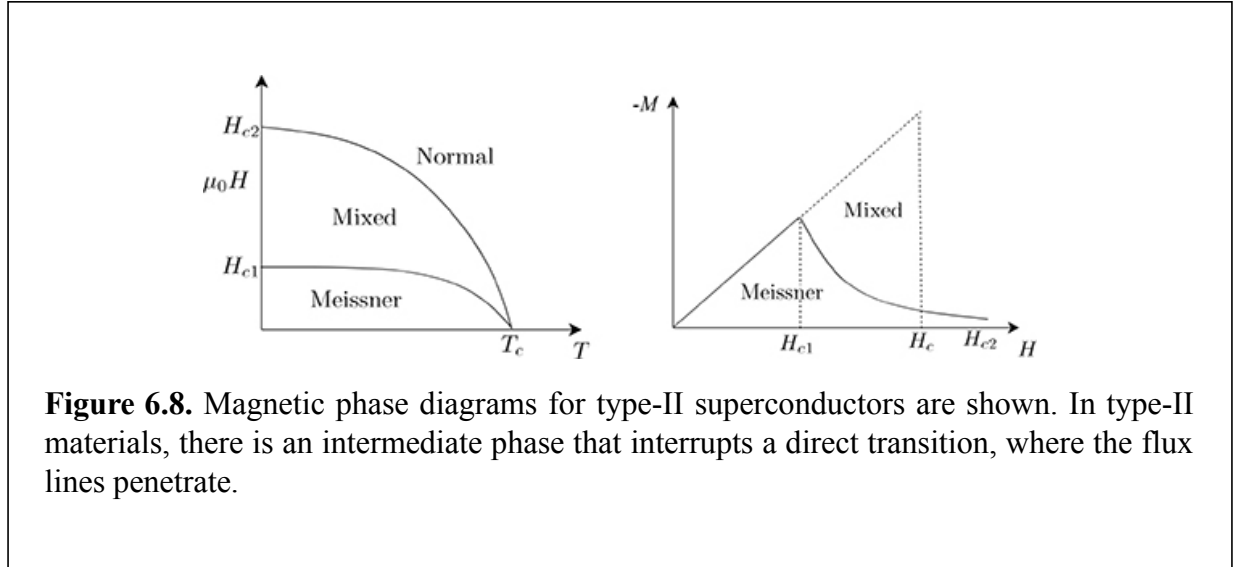


Figure 6.8. Magnetic phase diagrams for type-II superconductors are shown. In type-II materials, there is an intermediate phase that interrupts a direct transition, where the flux lines penetrate.

From here we get a characteristic length called coherence length ξ . ξ is defined as the length scale over which the wavefunction or the order parameter does not vary much. The ratio of these two characteristic length scales is given by,

$$\kappa = \frac{\lambda}{\xi}. \quad (6.25)$$

For typical superconductors, $\lambda = 500\text{\AA}$, $\xi \approx 5000\text{\AA}$. Thus $\kappa < 1$.

In 1957, Abrikosov found that for some class of superconductors $\kappa > 1$. This is used as a distinguishing criterion for the type I and type II superconductors, in the following sense.

$$\begin{aligned} \kappa < 1 & \quad \text{type I superconductors} \\ \kappa > 1 & \quad \text{type II superconductors.} \end{aligned}$$

For type II superconductors, the flux lines penetrate the sample till a certain threshold magnetic field. A threshold value of κ for which such flux penetration starts to occur is $\kappa = \kappa_c \approx \frac{1}{\sqrt{2}}$. At this value of κ , the flux penetration starts at lower critical field H_{c1} and reaches $B_{\text{ext}} = H_{\text{ext}}$ at $H_{\text{ext}} = H_{c2}$.

Even experiments on the high- T_c superconductors confirmed many of the features in the magnetic phase diagram as discussed above. For example, the quantization of vortices ($\Phi_0 = \frac{h}{2e}$), Abrikosov phase, etc are observed in experiments. In the type-II superconductors, the identification of the upper critical field has particularly been controversial, quite unlike the conventional superconductors, where the onset of strong diamagnetism and superconducting

critical currents occur at the same H_{c2} , whereas in the high- T_c superconductors, they are found to occur at quite different values. There are several other features which are distinct, possibly because of defect induced pinning mechanism, etc. We shall not continue this discussion here any further, and will briefly come back to it at the end of the chapter. However, certain contrasting properties of these superconductors compared to the conventional ones will be mentioned during the course of discussion.

6.2.1 Thermodynamics of superconductors

The thermodynamics route is usually the easiest route to understand phase transition. Consider Gibbs free energy of a superconductor. If its magnetization is M and magnetic induction H , then the work done in bringing the superconductor into a region with magnetic induction H_{ext} from far away (where $H_{\text{ext}} = 0$) is given by,

$$W = -\mu_0 \int_0^{H_{\text{ext}}} M dH = \mu_0 \frac{H_{\text{ext}}^2}{2} \quad (6.26)$$

as $M = -H$ for a superconductor.

This is the extra Gibbs free energy of a superconductor. Let us call this g_s (per unit volume)

$$g_s(T, H_{\text{ext}}) = g_s(T, 0) + \frac{\mu_0 H_{\text{ext}}^2}{2} \quad (6.27)$$

or,

$$g_s(T, B_{\text{ext}}) = g_s(T, 0) + \frac{B_{\text{ext}}^2}{2\mu_0} \quad (6.28)$$

In the normal state, the magnetization is a very small value, thus the magnetic work term is vanishingly small. Hence

$$g_n(T, H_{\text{ext}}) = g_n(T, 0) \quad (6.29)$$

$g_s(T, H_{\text{ext}})$ increases quadratically with H_{ext} . For some H_{ext} , the normal and superconducting states have the same free energy. Beyond this critical value, the normal state is more stable. Equating equations (6.27) and 6.29 at $H_{\text{ext}} = H_c$ (see figure 6.9)

$$g_n(T, 0) = g_s(T, 0) + \frac{\mu_0 H_c^2}{2}. \quad (6.30)$$

The RHS is positive, and hence a superconducting state is more stable below H_c . For example, $H_c = 0.08$ T for Pb at $T = 0$ [7]. Thus at $T = 0$, the superconducting state is stabilized by 4.25×10^{-25} Joule/mole. This is a really small amount, however, interestingly this small number stabilizes the superconducting state.

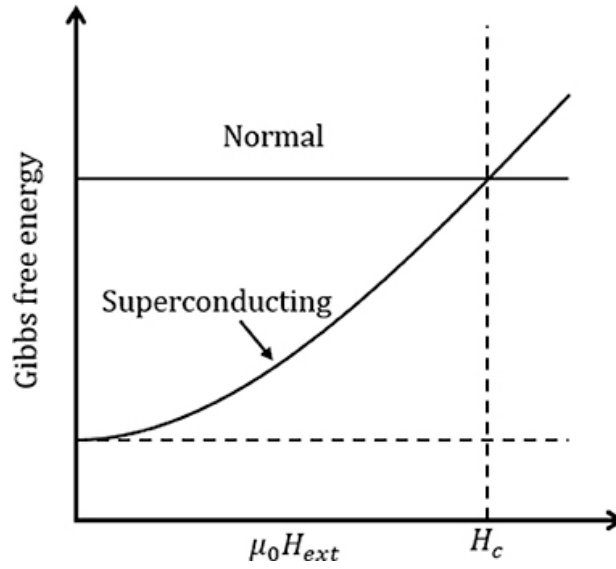


Figure 6.9. The Gibbs free energy is indicated schematically as a function of the external magnetic field. The energies for the superconducting and the normal states intersect at some value of the magnetic field, beyond which the normal state stabilizes.

6.2.2 Specific heat

The study of specific heat is in general interesting because it provides a good measure of the range of applicability of phonon mediated superconductivity. Above the transition temperature, the specific heat, C_n follows Debye's theory. In general for a normal metal, at temperatures $T < \theta_D$, the specific heat consists of:

- (i) electronic contribution, γT ;
- (ii) phonon contribution, AT^3 ;
- (iii) Schottky contribution, a/T^2 .

Schottky contribution can mostly be neglected and hence,

$$C_n = \gamma T + AT^3. \quad (6.31)$$

Thus, $C_n/T = \gamma + AT^2$. C_n/T versus T^2 thus denotes a linear plot.

At $T = T_c$, the contribution of electronic and phonon specific heat is,

$$\frac{C_{ph}}{C_e} = \frac{1}{\beta} \left(\frac{AT_c^2}{\gamma} \right)$$

where β is the ratio of the number of conduction electrons to that of atoms. From free-electron theory,

$$A = 234 \frac{R}{\theta_D^3}.$$

[8] Remember,

$$C_{ph} = \frac{12\pi^4}{5} R \left(\frac{T}{\theta_D} \right)^3 = 234 R \left(\frac{T}{\theta_D} \right)^3$$

Whereas,

$$\gamma = 4.93 \frac{R}{T_F}$$

and,

$$C_e = \frac{1}{2} \pi^2 R \left(\frac{T}{T_F} \right) = 4.93 R \left(\frac{T}{T_F} \right).$$

Thus,

$$\frac{C_{ph}}{C_e} = \left(\frac{47.5}{\beta} \cdot \frac{T_F}{\theta_D^3} \right) T_c^2$$

β can be set to be of the order of 1. For most of the conventional superconductors, $T_c \ll \theta_D$ So $C_e \gg C_{ph}$.

If the conduction electrons have effective mass, m^* , which is defined by the inverse of the curvature of the band structure, and that differs from bare mass m , then the conduction electron specific heat is given by,

$$\gamma = \left(\frac{m^*}{m} \right) \gamma_0$$

where γ_0 is the bare electron counterpart of γ .

$$\gamma_0 = \frac{\pi^2 R}{2T_F}.$$

Thus

$$\frac{m^*}{m} = \frac{\gamma}{\gamma_0} = \frac{2\gamma T_F}{\pi^2 R}.$$

It is worth mentioning here that for a typical high temperature superconductor, T_F is pretty low, $T_F \sim 4000$ K, at least an order lower than conventional elemental superconductors, such

as for Cu, $T_F \sim 80\,000$ K. The discrepancy is probably due to the large values of the effective masses.

With C_{es} numerically obtained, we can integrate to find the change in internal energy $U(T)$ as we decrease temperature from T_c . At $T = T_c$ it must be the same as the normal value $U_{\text{en}}(0) + \frac{1}{2}\gamma T_c^2$. So

$$U_{\text{es}}(T) = U_{\text{en}}(0) + \frac{1}{2}\gamma T_c^2 - \int_T^{T_c} C_{\text{es}} dT.$$

From the internal energy and entropy we may compute the free energy using,

$$F_{\text{es}}(T) = U_{\text{es}}(T) - TS_{\text{es}}(T).$$

The critical field is determined by

$$\mu_0 \frac{H_c^2(T)}{2} = F_{\text{en}}(T) - F_{\text{es}}(T)$$

where,

$$F_{\text{en}}(T) = U_{\text{en}}(0) - \frac{1}{2}\gamma T^2.$$

To compute the thermodynamic quantities, we explicitly consider the temperature dependence of the specific heat of a superconductor, and explore its behaviour at the normal-superconductor phase transition. There is a jump in the specific heat at $T = T_c$ as we have seen earlier.

The Gibb's free energy per unit volume for a superconductor in a magnetic field is written as,

$$G = U - TS - HM \tag{6.32}$$

neglecting the usual PV term. Since,

$$dU = TdS + HdM. \tag{6.33}$$

This is the change of internal energy in the presence of a magnetic field, from (6.32) and (6.33),

$$dG = -SdT - MdH. \tag{6.34}$$

Thus integrating (6.34),

$$G_s(H) = G_s(0) + \frac{H^2}{2}. \tag{6.35}$$

Along the critical curve where the SC and the normal metal are in equilibrium,

$$G_n = G_s(0) + \frac{H^2}{2}. \quad (6.36)$$

Where G_s and G_n are the Gibb's free energy for the superconducting and the normal states. From equation (6.34)

$$\left(\frac{\partial G}{\partial T} \right)_H = -S. \quad (6.37)$$

Thus, at equilibrium,

$$S_n - S_s = -H_c \frac{dH_c}{dT}. \quad (6.38)$$

S_s denotes the entropy of an SC in zero field. Since $\frac{dH_c}{dT}$ is always negative, the entropy of the normal state is always greater than that of the superconducting state. Thus,

$$\begin{aligned} \Delta C = C_s - C_n &= -T \frac{d}{dT} (S_s - S_n) \\ &= TH_c \frac{d^2 H_c}{dT^2} + T \left(\frac{dH_c}{dT} \right)^2. \end{aligned} \quad (6.39)$$

At $T \rightarrow T_c$, $H_c \rightarrow 0$,

$$\Delta C = T_c \left(\frac{dH_c}{dT} \right)^2. \quad (6.40)$$

From (6.38) at $T = T_c$, $H_c = 0$, there is no latent heat of transition $\Delta S = 0$, but according to (6.40) $\Delta C = \text{finite}$. So the transition is second order (at $T = T_c$). However, away from T_c , the phase transition has a latent heat and is a first-order phase transition. The behaviour of entropy (S), specific heat (C) (scaled by γT_c), internal energy (U) and Helmholtz free energy (F) (scaled by γT_c^2) are shown as a function of reduced temperature (T/T_c) are shown in figures 6.10 and 6.11). The discontinuity of C at $T = T_c$ is a noteworthy feature, and denotes a signature of a superconducting phase transition.

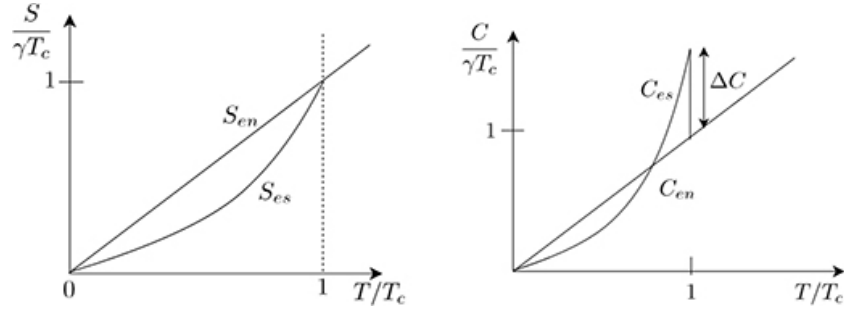


Figure 6.10. Schematic plots of entropy (S) and specific heat (C) scaled by γT_c (γ defined in text) are presented as a function of reduced temperature, $\frac{T}{T_c}$. $T/T_c < 1$ denotes the superconducting state, while $T/T_c > 1$ depicts the normal state.

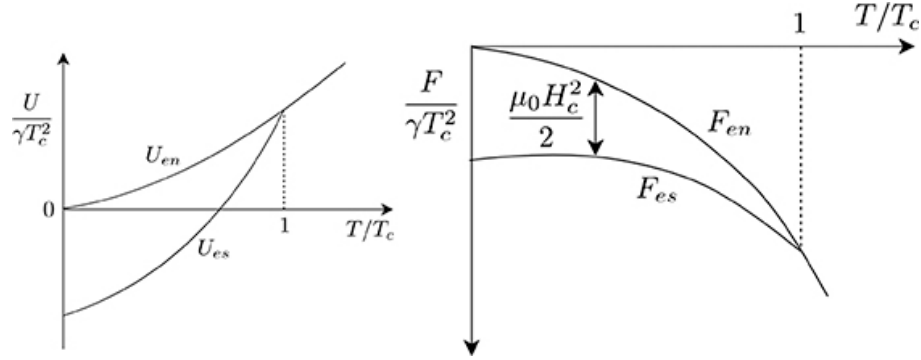


Figure 6.11. Schematic plots of entropy the internal energy and the Helmholtz free energy scaled by γT_c^2 are presented as a function of reduced temperature.

However, this is not true for high temperature superconductors. In fact for ($\text{La}_{0.9}\text{Sr}_{0.1}\text{CuO}_{4-\delta}$) and ($\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$) $AT_c^3 \gg \gamma$, so the phonon term dominates at $T = T_c$. The distinct behaviour of the specific heat for conventional and high- T_c superconductors as a function of temperature is shown in figure 6.12.

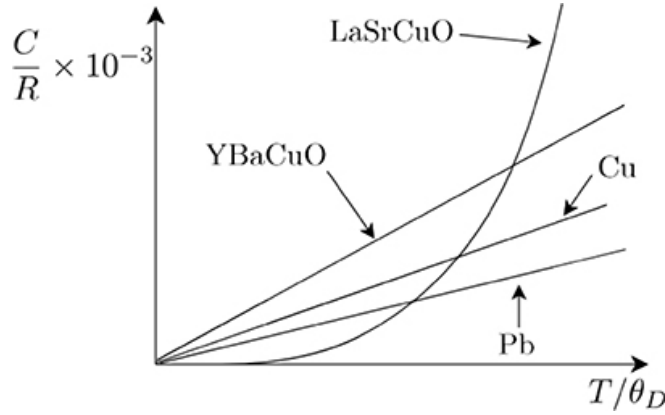


Figure 6.12. The behaviour of the specific heat is schematically shown as a function of the reduced temperature T/θ_D for both conventional (Cu and Pb) and unconventional (cuprate) superconductors.

6.2.3 Density of states

The Bogoliubov quasiparticle spectrum, E_k is easily seen to have a minimum of Δ_k for $\xi_k = 0$ (ξ_k is positive definite including zero). Thus, in addition to playing the role of the order parameter Δ_k is also the energy gap in the single-particle spectrum. To see this explicitly, we can do a change of variables from ξ_k (for the normal state energies) to E_k (denoted as the quasiparticle energies), namely,

$$N(E)dE \rightarrow N_n(\xi)d\xi$$

Since, the density of states (DOS) should remain conserved, as the number of carriers remains conserved,

$$N(E)dE = N_n(\xi)d\xi. \quad (6.41)$$

Since, the gap Δ is much smaller than the energy range ($\Delta \ll E$) over which the normal state DOS varies, we can replace,

$$N_N(\xi) \approx N_n(0) = N_0 \quad (6.42)$$

Now the quasiparticle energies are given by $E_k^2 = \Delta^2 + \xi_k^2$.

Dropping the subscripts,

$$E^2 = \Delta^2 + \xi^2$$

$$2EdE = 2\xi d\xi$$

$$\frac{d\xi}{dE} = \frac{E}{\xi} = \frac{E}{\sqrt{E^2 - \Delta^2}}.$$

Here,

$$N(E)dE = N_0 d\xi$$

$$\frac{N(E)}{N_0} = \frac{d\xi}{dE}.$$

Thus,

$$\frac{N(E)}{N(0)} = \begin{cases} \frac{E}{\sqrt{E^2 - \Delta^2}} & \text{for } E > \Delta \\ 0 & \text{for } E < \Delta. \end{cases}$$

Excitations with all momenta k , even those whose ξ_k fall in the gap have their energies raised above Δ . Moreover, we get a divergent DOS for $E \rightarrow \Delta$ from above.

Considering that across the phase transition, the DOS remains constant, i.e.

$$N_s(E)dE = N_n(\xi)d\xi.$$

Since we are interested in energy range of a few meV from the Fermi energy, we take $N_n(\xi) = N(\varepsilon_F) = \text{a constant}$.

$$\frac{N_s(E)}{N(\varepsilon_F)} = \frac{d\xi}{dE} = \frac{E}{\sqrt{E^2 - \Delta^2}} \quad \text{for } E > \Delta$$

The behaviour of the DOS is plotted in figure 6.13. It is zero for $E < \Delta$, and falls off as E becomes larger than the gap, Δ . The quasiparticle energies (E_k) as a function of the single-particle energies (ξ_k) are plotted in figure 6.14 where the superconducting phase exists above the parabola, and the dotted lines denote the boundaries separating the superconducting and the normal phases.

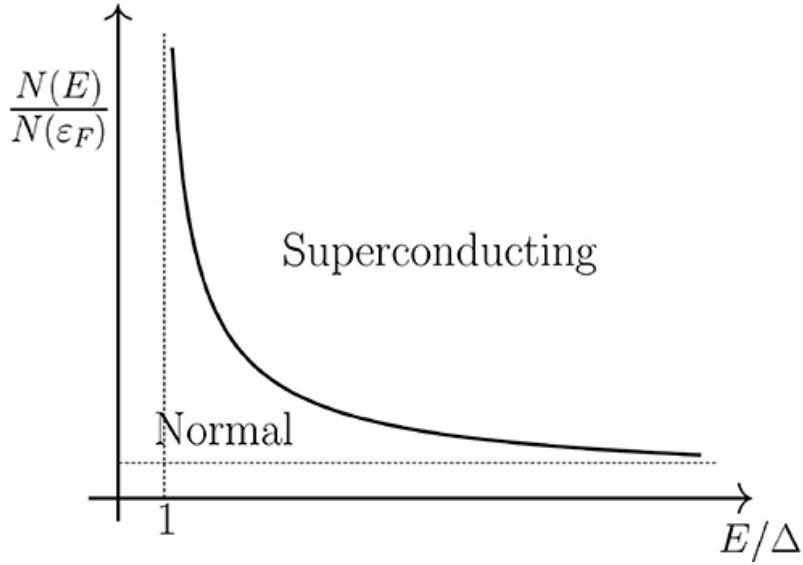


Figure 6.13. The DOS as a function of the scaled energy, E/Δ is shown. The right of the bold line denotes the superconducting phase, while on the left, it is the normal state.

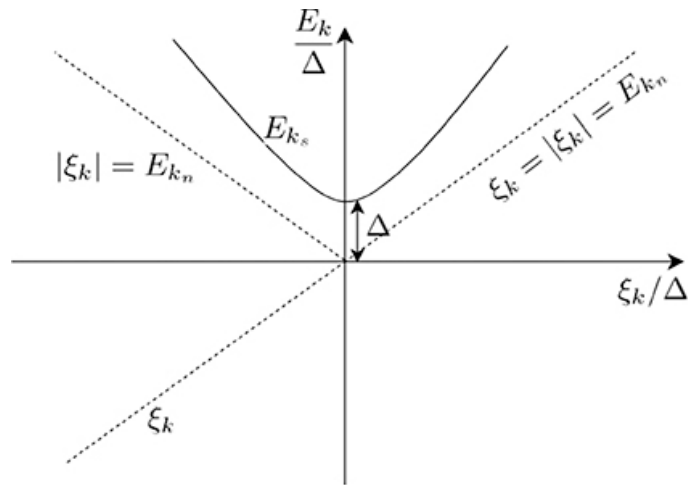


Figure 6.14. The scaled quasiparticle energy, E/Δ is plotted as function of scaled single-particle energy, ξ_k/Δ . The region above the parabola is superconducting (SC) and the region below the dotted lines denote metallic phases.

6.3 BCS theory

6.3.1 Introduction

Even though a lot of experimental progress occurred in the field of superconductivity, a complete understanding of the pairing phenomena and consequently the formation of the superconducting state were lacking. In the following, we describe a microscopic theory that addresses all of these concerns. The BCS theory of superconductivity was formulated by Bardeen, Cooper and Schrieffer in 1957. It accurately describes many of the properties of the weak coupling superconductors. The term ‘*weakly coupled*’ refers to the strength of the electron–phonon coupling that is considered as the origin of pairing in this class of superconductors, as we shall see below. The basic idea of the BCS theory is that the electrons in metal form bound pairs below a certain critical temperature, called the transition temperature. Not all the electrons take part in this pairing process, only the ones that reside close to the Fermi surface, and within a distance of the Debye energy, ω_D . The paired state of all of these electrons needs a many-body description.

6.3.2 Isotope effect

Fröhlich in 1950 [9] realized that the electrons could interact by exchanging phonons, and in such a process, the interaction could be attractive. That was the first proposal that superconductivity can originate from electron–phonon interaction. But how do the phonons come into the picture? The dependence on phonon parameters was experimentally demonstrated by the fact that the transition temperature, T_c is a function of the ionic mass for different isotopes of the same metal (that becomes a superconductor at lower temperatures). The effect of T_c is shown to have the form,

$$\Delta T_c / T_c = -\frac{1}{2} \left(\frac{\Delta M}{M} \right) \quad (6.43)$$

where M denotes the ionic mass, and ΔM is the difference in masses between the isotopes. This is called the isotope effect. It played a significant role in unravelling the issue concerning the attractive interaction among the charge carriers that eventually leads to superconductivity. Thus, we shall see that the contribution of the lattice excitations or the phonons is central to the formation of bound pairs. It will be shown later that physical quantities, such as the energy of the bound pairs, T_c , etc involve phonon frequencies, and thus should display isotope effect.

The isotope effect is more conveniently represented in terms of the isotope coefficient, α via the relation,

$$T_c \sim M^{-\alpha} \quad (6.44)$$

where α is given by,

$$\alpha = -\frac{M}{\Delta M} \frac{\Delta T_c}{T_c}. \quad (6.45)$$

In general, α has a value 0.5, however, it may vary from one material to another.

6.3.3 Origin of attractive interaction

We consider interaction between two electrons, one with $(\mathbf{k}, \uparrow)$ and the other $(-\mathbf{k}, \downarrow)$. One may consider a spin independent interaction, but since it is not relevant to the context, we can neglect the spin indices for future discussions. There are two mechanisms by which two electrons can interact: they can interact directly or via a third party, which in this case is phonons. We shall explore these two cases in the following.

Case I

One obvious process is a direct electron–electron interaction mediated via the Coulomb forces. Let us consider the initial state as $|i\rangle$ (before scattering) and the final state $|f\rangle$ after scattering. Let us denote this interaction term as H_{e-e}^{dir} . The matrix elements of such a term between the states can be written as,

$$\langle i | H_{e-e}^{\text{dir}} | f \rangle = \int e^{i\mathbf{k}\cdot\mathbf{r}} U_c(\mathbf{r}) e^{-i\mathbf{k}'\cdot\mathbf{r}} d^3\mathbf{r}. \quad (6.46)$$

Assuming both $|i\rangle$ and $|f\rangle$ are plane wave states. This term cannot yield a negative contribution and hence does not contribute to the formation of a bound state. The energies involved in this case can be enumerated as:

- (i) initial: $\epsilon_i = 2\xi_{\mathbf{k}} \quad \xi_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} - \mu;$
- (ii) final: $\epsilon_f = 2\xi_{\mathbf{k}'}, \quad \mathbf{k}' = \mathbf{k} + \mathbf{q}.$

Case II

Now let us consider an indirect process. Consider scattering of two electrons via exchange of a phonon. Thus there is an intermediate process involved, and hence it is a second-order process. There are two ways that such a second-order process can occur:

- (i) electron 1 with momentum $\mathbf{k}$ emits a phonon, of frequency ω_q which is later re-absorbed by electron 2 with momentum $-\mathbf{k}$ (see left panel of figure 6.15);
- (ii) electron 2 with momentum $-\mathbf{k}$ emits a phonon with frequency ω_q which is later re-absorbed by electron 1 with momentum $\mathbf{k}$ (see right panel of figure 6.15).

Since both these processes are equally probable, they need to be incorporated with equal weight. The corresponding energies involved in the second case are:

- (i) initial state (before scattering): $\epsilon_i = 2\xi_{\mathbf{k}};$
- (ii) final state (after scattering): $\epsilon_f = 2\xi_{\mathbf{k}'},$

Even if the initial and the final states are the same as in case I, the intermediate state is different. However, the corresponding energies are still the same, namely,

$$\begin{aligned} \epsilon_{\text{int}}^{(i)} &= \xi_{\mathbf{k}'} + \xi_{\mathbf{k}} + \hbar\omega_q \\ \epsilon_{\text{int}}^{(ii)} &= \xi_{\mathbf{k}} + \xi_{\mathbf{k}'} + \hbar\omega_q. \end{aligned}$$

Now the matrix element of $H_{e-e}^{\text{indirect}}$ can be calculated as,

$$\begin{aligned}\langle i | H_{e-e}^{\text{indirect}} | f \rangle &= \sum_{\text{int}} \frac{\langle i | H_{e-e}^{\text{indirect}} | \text{int} \rangle \langle \text{int} | H_{e-e}^{\text{indirect}} | f \rangle}{(E_{i,f} - E_{\text{int}})^2} \\ &= \sum_{\text{int}} \langle i | H_{e-e}^{\text{indirect}} | \text{int} \rangle \frac{1}{2} \left[\frac{1}{E_f - E_{\text{int}}} - \frac{1}{E_i - E_{\text{int}}} \right] \\ &\quad \times \langle \text{int} | H_{e-e}^{\text{indirect}} | f \rangle\end{aligned}\tag{6.47}$$

where the summation is over the intermediate states. Note that the energy denominators can be written as,

$$\begin{aligned}\frac{1}{E_f - E_{\text{int}}} &= \frac{1}{2\xi_{\mathbf{k}'} - \xi_{\mathbf{k}} - \xi_{\mathbf{k}'} - \hbar\omega_q} \\ \frac{1}{E_i - E_{\text{int}}} &= \frac{1}{2\xi_{\mathbf{k}} - \xi_{\mathbf{k}'} - \xi_{\mathbf{k}} - \hbar\omega_q}\end{aligned}\tag{6.48}$$

where, $\xi_{\mathbf{k}'} - \xi_{\mathbf{k}} = \hbar\omega$. Thus,

$$\begin{aligned}\langle i | H_{e-e}^{\text{indirect}} | f \rangle &= \frac{1}{\omega - \omega_q} - \frac{1}{\omega + \omega_q} | \widetilde{V}_c(q) |^2 \\ &= \frac{2\omega_q}{\omega^2 - \omega_q^2} | \widetilde{V}_c(q) |^2.\end{aligned}\tag{6.49}$$

The matrix elements are positive definite, so we are interested to know the sign of the energy denominator. For $\omega < \omega_q$, it is negative, and hence attractive. This is known as Cooper's instability. In real physical situations, the Coulomb term gets weakened by screening, which eventually takes a form,

$$V(\mathbf{q}, \omega) = \frac{4\pi e^2}{q^2 + k_0^2} + \frac{4\pi e^2}{q^2 + k_0^2} \frac{\omega_q}{\omega^2 - \omega_q^2}.\tag{6.50}$$

A schematic plot of the interaction term as a function of the electron energy, ω is plotted in figure 6.16. In a small region, between the points A and B marked in figure 6.16, $V(\mathbf{q}, \omega)$ becomes negative and hence attractive.

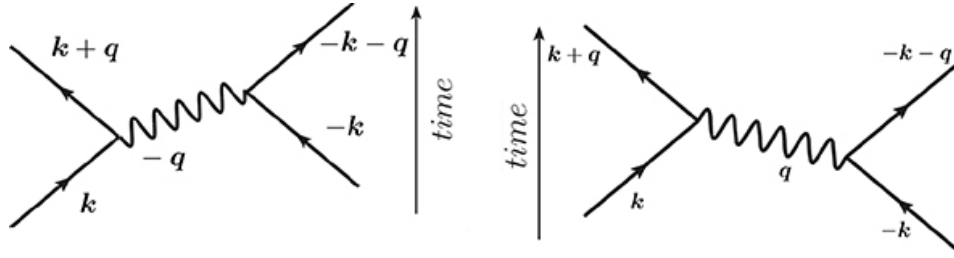


Figure 6.15. Feynman diagrams are shown for the electron–electron interaction mediated via phonons. The left panel shows electron 1 emitting a phonon which is captured by electron 2 at a later time. The right panel shows electron 2 emitting a phonon which is captured by electron 1 at a later time.

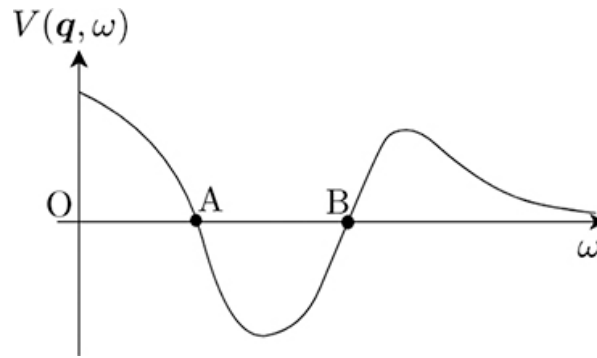


Figure 6.16. A schematic plot of the interaction energy $V(\mathbf{q}, \omega)$ plotted as a function of the electron energy ω . The region between A and B is where the interaction becomes attractive.

6.3.4 The BCS ground state

The ground states of a free Fermi gas correspond to the filled states with energy ϵ below ϵ_F and all states empty beyond $\epsilon > \epsilon_F$. This Fermi gas of electrons becomes unstable against the formation of at least one bound pair, regardless of how weak the interaction is, so long as it is attractive [10].

It is well known that the binding does not ordinarily occur in a two-body problem in two or three dimensions, until the strength of the potential exceeds a finite threshold value [11]. So, in order to see how this binding comes about, we consider a simple model of two electrons added to the Fermi sea at $T = 0$. The restriction imposed here is that these two electrons interact with

each other, but not with those inside the Fermi sea, except, of course due to the exclusion principle.

Thus, we seek a two-particle wavefunction for our purpose. Bloch argued that the lowest energy state for a two-particle system has zero total momentum. Thus, the two electrons have equal and opposite momenta. This means the orbital wavefunction, $\psi_0(\mathbf{r}_1, \mathbf{r}_2)$ should be of the form,

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) = \sum_{\mathbf{k}} g_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_1} e^{-i\mathbf{k} \cdot \mathbf{r}_2}. \quad (6.51)$$

Taking into account the asymmetry of the total wavefunction with respect to the exchange of two electrons, ψ_0 contains even functions, such as, $\cos[\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)]$ with an antisymmetric spin part of the form, $[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$, or a sum of products of $\sin[\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)]$ with a symmetric triplet function of the form, $[|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle]$ and $[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle]$. In a two-particle system, we expect the singlet to have lower energy because the cosinusoidal dependence of the orbital wavefunction $\mathbf{r}_1 - \mathbf{r}_2$ gives a larger probability for the electrons to be near to each other. Thus, we zero in on a two-electron singlet wavefunction of the form,

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) = \left[\sum_{\mathbf{k} > k_F} g_{\mathbf{k}} \cos[\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)] \right] [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]. \quad (6.52)$$

This can be plugged into the Schrödinger equation to yield,

$$(E - 2\epsilon_{\mathbf{k}})g_{\mathbf{k}} = \sum_{\mathbf{k}' > k_F} V_{\mathbf{k}\mathbf{k}'} g_{\mathbf{k}'}. \quad (6.53)$$

The above equation is true for any value of $\mathbf{k}$. $\epsilon_{\mathbf{k}}$ are the single-particle energies (of the plane wave type), and $V_{\mathbf{k}\mathbf{k}'}$ are the matrix elements of the interaction potential,

$$V_{\mathbf{k}\mathbf{k}'} = \frac{1}{V} \int V(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} d\mathbf{r} \quad (6.54)$$

where $\mathbf{r}$ denotes the spatial distance between the two electrons. $V_{\mathbf{k}\mathbf{k}'}$ describes a scattering of a pair of electrons with momenta $(\mathbf{k}'_{\uparrow}, -\mathbf{k}'_{\downarrow})$ to $(\mathbf{k}_{\uparrow}, -\mathbf{k}_{\downarrow})$. We have to solve for the amplitude, $g_{\mathbf{k}}$ such that the total energy $E < 2\epsilon_F$, such that a bound pair exists.

It is hard to carry out calculations for a general form of $V_{\mathbf{k}\mathbf{k}'}$. Cooper introduced a notable simplification,

$$\begin{aligned} V_{\mathbf{k}\mathbf{k}'} &= -V \quad \text{for} \quad \epsilon < \epsilon_{\mathbf{k},\mathbf{k}'} < \epsilon_F + \hbar\omega_D \\ &= 0 \quad \text{otherwise} \end{aligned} \quad (6.55)$$

where V denotes the strength of the electron–electron interaction. This implies,

$$g_{\mathbf{k}} = V \sum_{\mathbf{k}'} \frac{g_{\mathbf{k}'}}{2\epsilon_{\mathbf{k}} - E} \quad (6.56)$$

summing over the momentum index k ,

$$\frac{1}{V} = \sum_{k > k_F} \frac{1}{2\epsilon_k - E}. \quad (6.57)$$

Note that this summation over k is for k values that are larger than the Fermi wavevector. We replace the summation by an integration, via using the DOS at the Fermi level, namely, $N(\epsilon_F)$, which is constant for a particular material, and hence can be brought out of the integral. Hence,

$$\frac{1}{V} = N(\epsilon_F) \int_{\epsilon_F}^{\epsilon_F + \hbar\omega_D} \frac{d\epsilon}{2\epsilon - E} = \frac{1}{2} N(\epsilon_F) \ln \left(\frac{2\epsilon_F - E + 2\hbar\omega_D}{2\epsilon_F - E} \right). \quad (6.58)$$

For conventional superconductors,

$$N(\epsilon_F)V < 0.3.$$

This allows applicability of weak coupling approximation valid for $N(\epsilon_F)V \ll 1$. This yields,

$$\begin{aligned} \frac{2}{N(\epsilon_F)V} &= \ln \left(\frac{2\epsilon_F - E + 2\hbar\omega_D}{2\epsilon_F - E} \right) \\ \text{or, } \frac{2\epsilon_F - E + 2\hbar\omega_D}{2\epsilon_F - E} &= e^{2/N(\epsilon_F)V} \\ 2\epsilon_F - E + 2\hbar\omega_D &= (2\epsilon_F - E)e^{2/N(\epsilon_F)V} \\ E &= 2\epsilon_F - 2\hbar\omega_D e^{-2/N(\epsilon_F)V}. \end{aligned} \quad (6.59)$$

This is the expression for the total energy of the system of two electrons in the vicinity of a Fermi sea. Thus with respect to the Fermi surface of two electrons, the energy is negative, and hence it corresponds to a bound state. A bound state of two electrons is thus formed even for an infinitesimal interaction strength. Also, the form says that the binding energy is not analytic at $V = 0$, so it cannot be expanded in powers of V . Thus, any finite order perturbation theory would not be able to yield this result, and thus the theory is non-perturbative.

As for the wavefunction, it is dependent on the relative coordinate, $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, and is proportional to,

$$\sum_{k > k'} \frac{\cos(\mathbf{k} \cdot \mathbf{r})}{2\xi_{\mathbf{k}} + E'}$$

where, $\xi_{\mathbf{k}} = \epsilon_k - \epsilon_F$, and $E' = 2\epsilon_F - E > 0$. E' can now be called the binding energy relative to $2\epsilon_F$. A few comments are in order.

- (i) The amplitude of the wavefunction, namely, $(2\xi_k + E)^{-1}$ has its maximum value $1/E$ when $\xi_k = 0$, that is, for electrons at the Fermi level. It falls off at positive values of ξ_k . Thus, it elucidates the dominant role of the electrons at the Fermi energy on the pairing mechanism.
- (ii) The electronic states within an energy range E about ϵ_F are involved in the formation of the bound state.
- (iii) Note that $E < \hbar\omega_D$ for $N(\epsilon_F)V < 1$, this makes sure that the detailed behaviour of $V_{kk'}$ is not important, and the assumption of the form in equation (6.55) will suffice.
- (iv) Also, the small energy range allows estimation of the range of the Cooper pairs via uncertainty principle, namely $\xi_0 = \frac{\hbar v_F}{kT_c}$. Thus the pairs are highly overlapping.

6.3.5 Statistical description of the BCS ground state

What we have learnt so far is that the filled Fermi sea becomes unstable to the formation of a bound pair. In s -wave superconductivity, a large number of bound pairs form and a system condenses into ground state below the same transition temperature, T_c . Let us postulate a paired many-body ground state of the form (since it is a ground state, we assume it to be a singlet state),

$$|\psi_0\rangle = \sum_{k > k_F} g_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger |0\rangle \quad (6.60)$$

where $|0\rangle$ does not denote vacuum, rather it represents a filled Fermi sea. Suppose there are M electrons, and we have to involve N electrons of those to form $\frac{N}{2}$ pairs, the number of ways $\frac{N}{2}$ pairs can be formed from M electrons is,

$$\frac{M!}{\left(M - \frac{N}{2}\right)! \left(\frac{N}{2}\right)!} \approx (10^{20})^{20} \quad \text{for } M = 10^{23}. \quad (6.61)$$

Thus a solution of the problem demands that $(10^{20})^{20}$ number of g_k be solved, which is certainly an impossible task. Thus a statistical description must be employed here. Further, since the number of particles is large, no serious error will be done if one works with grand canonical ensemble, where only the average number of particles, that is, $\bar{N}$ is fixed.

To circumvent a significantly difficult problem, the BCS theory postulated a ground state of the form,

$$|\psi_G\rangle = \prod_{k=k_1, k_2, \dots, k_M} (u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger |\psi_0\rangle). \quad (6.62)$$

The probability that a paired state $(k_\uparrow, -k_\downarrow)$ is occupied is denoted by $|v_k|^2$, while it is unoccupied is given by $|u_k|^2$ with the condition,

$$|u_k|^2 + |v_k|^2 = 1 \quad \forall k. \quad (6.63)$$

Take an example of two states $\mathbf{k}_1$ and $\mathbf{k}_2$, then the wavefunction amplitudes represent,

$$\begin{aligned} u_{k_1}, u_{k_2} &\rightarrow 0 \text{ pair} \\ u_{k_1}, v_{k_2} &\rightarrow 1 \text{ pair in } (\mathbf{k}_2, -\mathbf{k}_2) \\ v_{k_1}, v_{k_2} &\rightarrow 2 \text{ pairs in } (\mathbf{k}_1, -\mathbf{k}_1), (\mathbf{k}_2, -\mathbf{k}_2). \end{aligned}$$

We can calculate the average number of particles in the following way,

$$\begin{aligned} \bar{N} &= \langle \hat{N}_{OP} \rangle = \left\langle \sum_{\mathbf{k}, \sigma} \hat{n}_{\mathbf{k}, \sigma} \right\rangle \\ &= \langle \psi_G | \sum_{\mathbf{k}} c_{k\uparrow}^\dagger c_{k\uparrow} + c_{k\downarrow}^\dagger c_{k\downarrow} | \psi_G \rangle \\ &= 2 \sum_k \langle \psi_G | c_{k\uparrow}^\dagger c_{k\uparrow} | \psi_G \rangle \\ &= 2 \sum_k \langle \psi_0 | (u_k^* + v_k^* c_{-k\downarrow}^\dagger c_{k\uparrow}^\dagger) c_{k\uparrow}^\dagger c_{k\uparrow} (u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) \\ &\quad \times \prod_{l \neq k} (u_l^* + v_l^* c_{-l\downarrow}^\dagger c_{l\uparrow}^\dagger) (u_l + v_l c_{l\uparrow}^\dagger c_{-l\downarrow}^\dagger) | \psi_0 \rangle. \end{aligned}$$

In principle u_k and v_k are complex quantities. We have also used $\langle A\phi | | \psi \rangle = \langle \phi | A^\dagger | \psi \rangle$. Let us look at various terms.

First, let us look at the term for $l \neq k$, namely,

$$|u_l|^2 + u_l^* v_l c_{l\uparrow}^\dagger c_{-l\downarrow}^\dagger + v_l^* u_l c_{-l\downarrow}^\dagger c_{l\uparrow}^\dagger + |v_l|^2 c_{-l\downarrow}^\dagger c_{l\uparrow}^\dagger c_{l\uparrow}^\dagger c_{-l\downarrow}^\dagger \quad (6.64)$$

where we have taken the expectation with $|\psi_0\rangle$. The middle two terms (second and the third terms) yield zero as they change occupancy of the state with l th pair (first term creates and the second term destroys). The last term only yields a normalization because it creates and destroys a pair. Thus the $l \neq k$ term yields,

$$|u_l|^2 + |v_l|^2$$

which, by normalization is equal to 1.

When the same procedure is applied to $l = k$ term, we shall have two terms surviving with $u_k v_k$ that will cancel out. Further,

$$|u_k|^2 c_{k\uparrow}^\dagger c_{k\uparrow} | \phi_0 \rangle = 0$$

since there are no states to annihilate for $k > k_F$. Thus we are left with only $|v_k|^2$.

Thus the average number of particles yields,

$$\bar{N} = 2 \sum_k |v_k|^2.$$

The factor 2 appears for the summation over spin. Further, our assumption that this reinforces $|v_k|^2$ indeed denotes probability of occupied states.

To calculate some more statistical quantities that may be of physical relevance, we estimate the RMS fluctuations of $\bar{N}$, we calculate,

$$\begin{aligned} \langle (N - \bar{N})^2 \rangle &= \langle N^2 - 2N\bar{N} + \bar{N}^2 \rangle \\ &= \langle N^2 \rangle - \bar{N}^2 \\ &= 4 \sum_k u_k^2 v_k^2. \end{aligned} \tag{6.65}$$

This is a positive definite quantity. In fact, v_k as a function of k goes from 1 to zero and u_k goes from $0 \rightarrow 1$ in an energy range given by $k_B T_c$. Thus the sum above is proportional to $\frac{T_c}{T_F} \bar{N}$

6.4 The variational calculation

We write down a generic Hamiltonian that encodes the kinetic energy and a two-particle interaction term as in the following,

$$\mathcal{H} = \sum_{k,\sigma} \xi_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma} + \sum_{k,l} V_{kl} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger c_{-l\downarrow} c_{l\uparrow} = \hat{K} + \hat{V} \tag{6.66}$$

where $\hat{K}$ and $\hat{V}$ denote the kinetic and the interaction energies, and $\xi_k = \epsilon_k - \mu$. k, l are the momenta indices. As exact solutions are hard to arrive at, and perturbation theory will not be of relevance for reasons discussed earlier, we shall resort to a variational calculation, that is, minimize the energy with respect to the tunable parameters, which are the coefficients u_k and v_k . That is,

$$\delta \left[\langle \psi_G | \sum_{k,\sigma} \xi_k n_{k\sigma} + \sum_{k,l} V_{kl} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger c_{-l\downarrow} c_{l\uparrow} | \psi_G \rangle \right]. \tag{6.67}$$

Let us look at the kinetic energy term. We have already computed it earlier, namely,

$$\langle \hat{K} \rangle = \langle \psi_G | \sum_{k,\sigma} \xi_k n_{k\sigma} | \psi_G \rangle = 2 \sum_k |v_k|^2 \xi_k. \tag{6.68}$$

Similarly for the potential energy term,

$$\langle \hat{V} \rangle = \sum_{k,l} V_{kl} u_k^* v_l^* u_l v_k$$

where we have used the normalization constraint, that is, $|u_k|^2 + |v_k|^2 = 1 \quad \forall \quad k$. (6.69)

Doing a variational calculation with two variables is a difficult job, which, however, is simplified owing to the normalization condition. This yields an identification of (u_k, v_k) with $(\sin \theta_k, \cos \theta_k)$ which facilitates the Hamiltonian to be dependent only on one parameter, namely θ_k . In particular, we take,

$$u_k = \sin \theta_k, v_k = \cos \theta_k. \quad (6.70)$$

Thus we can minimize energy with respect to θ_k , which implies,

$$\frac{\partial}{\partial \theta_k} \left[\sum_{k'} \xi_{k'} (1 + \cos 2\theta_{k'}) + \frac{1}{4} \sum_{k', l} V_{k'l} \sin 2\theta_{k'} \sin 2\theta_l \right] = 0 \quad (6.71)$$

$$-2\xi_k \sin 2\theta_k + \sum_l V_{kl} \cos 2\theta_k \sin 2\theta_l = 0. \quad (6.72)$$

Now define,

$$\Delta_k = -\frac{1}{2} \sum_l V_{kl} \sin 2\theta_l = -\frac{1}{2} \sum_l V_{kl} \sin 2\theta_l. \quad (6.73)$$

Using this in the above equation can be written as,

$$2\xi_k \sin 2\theta_k = \frac{1}{2} \sum_l V_{kl} \cos 2\theta_k \sin 2\theta_l = -2\Delta_k \cos 2\theta_k. \quad (6.74)$$

This yields,

$$\tan 2\theta_k = -\frac{\Delta_k}{\xi_k}, \quad \text{implying that,} \quad \frac{\sin 2\theta_k}{\cos 2\theta_k} = -\frac{\Delta_k}{\xi_k}. \quad (6.75)$$

Using the definitions,

$$\begin{aligned} 2u_k v_k &= \sin 2\theta_k = \frac{\Delta_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \\ v_k^2 - u_k^2 &= \cos 2\theta_k = -\frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}}. \end{aligned} \quad (6.76)$$

We have,

$$v_k^2 - u_k^2 = -\frac{\xi_k}{E_k}$$

where, $E_k = \sqrt{\xi_k^2 + \Delta^2}$, and, of course the normalization condition, namely, $u_k^2 + v_k^2 = 1$. Thus the definition for the coherence factors can be written as,

$$\begin{aligned} v_k^2 &= \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right) = \frac{1}{2} \left[1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right] \\ u_k^2 &= \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right) = 1 - v_k^2. \end{aligned} \quad (6.77)$$

This is a choice which seems to be fitting all definitions. There is a close resemblance between the v_k^2 for the BCS ground state at $T = 0$, and the free Fermi distribution at $T = T_F$ (see figure 6.17). Also, v_k^2 falls off as $1/\xi_k^2$ for $\xi_k \gg \Delta$. If one remembers, g_k^2 falls off in a similar fashion in the simple treatment of a single Cooper pair presented earlier. Also we have chosen $\cos 2\theta_k$ with a negative sign, because if ξ_k is large (that is, $\epsilon_k \gg \mu$) v_k should go to zero.

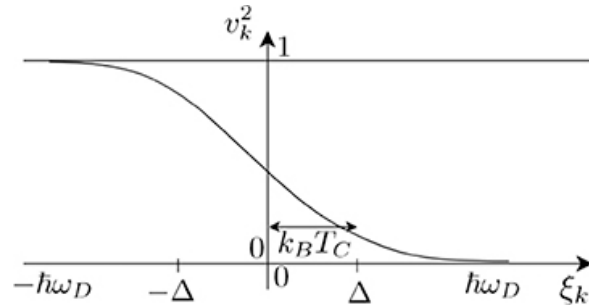


Figure 6.17. The behaviour of v_k^2 (probability of states occupied by a pair of momentum k and $-k$) is shown as a function of ξ_k ($= \epsilon_k - \mu$).

Hence, Δ_k assumes a form,

$$\begin{aligned} \Delta_k &= -\frac{1}{2} \sum_l V_{kl} \sin 2\theta_l \\ \Delta_k &= -\frac{1}{2} \sum_l V_{kl} \frac{\Delta_l^2}{\sqrt{\Delta_l^2 + \xi_l^2}}. \end{aligned} \quad (6.78)$$

A trivial solution of the above equation is $\Delta = 0$. This implies, $2u_kv_k = 0$, and $v_k^2 - u_k^2 = -1$. Since, $u_k^2 + v_k^2 = 1$, it implies that $v_k^2 = 0$, which in turn means a normal

state and hence there are no pairs. The above discussion clearly indicates that Δ_k can be used as order parameters for the superconducting transition, which when finite implies a superconducting state, and $\Delta_k = 0$ denotes a normal state.

Let us recall Cooper's original assumption on the nature of the interaction, that is,

$$V_{kl} = -V \quad \text{if} \quad |\xi_k - \xi_l| < \hbar\omega_D.$$

Hence putting it in equation (6.73),

$$\begin{aligned} \Delta_k &= \frac{1}{2}V \sum_l \sin 2\theta_l \quad \text{for} \quad |\xi_k - \xi_l| < \hbar\omega_D \\ &= 0 \quad \text{otherwise.} \end{aligned} \tag{6.79}$$

From the above equation, it is clear that the order parameter, Δ is independent of the momentum index, k . Substituting $\sin 2\theta_l$ from equation (6.76), one may cancel Δ from both sides to get,

$$\begin{aligned} 1 &= \frac{V}{2} \sum_l \frac{1}{\sqrt{\Delta^2 + \xi_l^2}} \\ \frac{1}{V} &= \int_0^{\hbar\omega_D} \frac{N(\varepsilon_F) d\xi}{2\sqrt{\Delta^2 + \xi^2}} \\ \frac{2}{N(\varepsilon_F)V} &= \int_0^{\hbar\omega_D} \frac{d\xi}{\sqrt{\Delta^2 + \xi^2}}. \end{aligned} \tag{6.80}$$

Integrating and solving for Δ ,

$$2 \sinh^{-1} \left(\frac{\xi}{\Delta} \right) \Big|_0^{\hbar\omega_D} = \frac{2}{N(\varepsilon_F)V} \tag{6.81}$$

$$\text{Or, } 2 \sinh^{-1} \left(\frac{\hbar\omega_D}{\Delta} \right) = \frac{2}{N(\varepsilon_F)V}. \tag{6.82}$$

Reorganizing,

$$\Delta = \frac{\hbar\omega}{\sinh(1/N(\varepsilon_F)V)}. \tag{6.83}$$

For a weak coupling superconductor, $N(\varepsilon_F)V \ll 1$, one arrives at,

$$\Delta = 2\hbar\omega_D e^{-1/N(\varepsilon_F)V}. \tag{6.84}$$

This is the expression for the energy difference between the superconducting state and the normal state. Thus this much of energy has to be supplied in order to break a superconducting

state, and achieve a normal (metallic) state. Suppose a thermal excitation facilitates such a transition, which yields, $\Delta \sim k_B T_c$; *a priori*, from BCS theory one gets (will be shown immediately afterwards),

$$k_B T_c \approx 1.13 \hbar \omega_D \exp[-1/N(\varepsilon_F)V] \quad (6.85)$$

which only changes the factor ‘2’ in equation (6.84) to 1.13 in equation (6.85). Having found Δ , we may simply compute the coefficients u_k and v_k which should specify the optimal BCS wavefunction.

6.4.1 Temperature dependence of the gap

We have introduced that the energy gap (which also acts as the order parameter) in a superconductor is defined by,

$$\Delta_k = - \sum_{k'} V_{kk'} \langle c_{-k'\downarrow} c_{k'\uparrow} \rangle \quad (6.86)$$

Δ_k allows us to write the mean field BCS Hamiltonian,

$$\mathcal{H} = \sum_{k,\sigma} \xi_k c_{k\sigma}^\dagger c_{k\sigma} - \sum_{k,\sigma} (\Delta_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger + \Delta_k^* c_{-k\downarrow} c_{k\uparrow}). \quad (6.87)$$

We have neglected a constant term in the above equation. This Hamiltonian can be diagonalized by Bogoliubov–Valatin transformation, where the creation and the annihilation operators are written in terms of the quasiparticle operators as,

$$c_{k\uparrow} = u_k^* \gamma_{k_0} + v_k^* \gamma_{k_1} \quad (6.88)$$

$$c_{-k\downarrow}^\dagger = -v_k^* \gamma_{k_0} + u_k^* \gamma_{k_1}. \quad (6.89)$$

Being a linear combination of c_k and $c_k^\dagger$, γ_k obeys fermionic anticommutator relations and thus the relationships are canonical. This along with the normalization condition yields,

$$\begin{aligned} \mathcal{H} = & \sum_k \xi_k [(|u_k|^2 - |v_k|^2)(\gamma_{k_0}^\dagger \gamma_{k_0} + \gamma_{k_1}^\dagger \gamma_{k_1}) + 2|v_k|^2 + 2u_k^* v_k^* \gamma_{k_1} \gamma_{k_0} + 2u_k v_k \gamma_{k_0}^\dagger \gamma_{k_0}^\dagger] \\ & + \sum_k (\Delta_k u_k v_k^* + \Delta_k^* u_k^* v_k)(\gamma_{k_0}^\dagger \gamma_{k_0} + \gamma_{k_1}^\dagger \gamma_{k_1} - 1) + (\Delta_k v_k^{*2} - \Delta_k^* u_k^{*2}) \gamma_{k_1} \gamma_{k_0} \\ & + (\Delta_k^* v_k^2 - \Delta_k u_k^2) \gamma_{k_0}^\dagger \gamma_{k_1}^\dagger. \end{aligned} \quad (6.90)$$

Finally, one gets,

$$\mathcal{H} = \sum_k E_k (\gamma_{k_0}^\dagger \gamma_{k_0} + \gamma_{k_1}^\dagger \gamma_{k_1}) + \sum_k (\xi_k - E_k + |\Delta_k|^2)$$

where $E_k = \sqrt{\xi_k^2 + |\Delta|^2}$. The second term is just a constant. So that the Hamiltonian is diagonal in the basis of the quasiparticle operators³. (6.91)

For the gap function, Δ , in terms of the quasiparticle operators after a little bit of algebra yields,

$$\Delta_k = - \sum_{k'} V_{kk'} u_{k'}^* v_{k'} \langle 1 - \gamma_{k_0}^\dagger \gamma_{k_0} - \gamma_{k_1}^\dagger \gamma_{k_1} \rangle. \quad (6.92)$$

At zero temperature, $\langle \gamma_{k_0}^\dagger \gamma_{k_0} \rangle = 1$, however, at non-zero temperatures,

$$\langle \gamma_{k_0}^\dagger \gamma_{k_0} \rangle = f(E_k) = \frac{1}{e^{\beta E_k} + 1}$$

where $f(E_k)$ denotes the Fermi distribution function for the quasiparticles. Since expectation value of each of the number operators yields a Fermi function, one gets,

$$\langle 1 - \gamma_{k_0}^\dagger \gamma_{k_0} - \gamma_{k_1}^\dagger \gamma_{k_1} \rangle = 1 - 2f(E_k).$$

Thus, for the gap function,

$$\begin{aligned} \Delta_k &= - \sum_{k'} V_{kk'} u_{k'}^* v_{k'} (1 - 2f(E_k)) \\ &= - \sum_{k'} V_{kk'} \frac{\Delta_{k'}}{2E_{k'}} \tanh\left(\frac{\beta E_{k'}}{2}\right). \end{aligned}$$

With $V_{kk'} = -V$,

$$\frac{1}{V} = \frac{1}{2} \sum_k \frac{\tanh(\beta E_k/2)}{E_k}. \quad (6.93)$$

The above equation holds a clue to the temperature dependence of the energy gap, $\Delta(T)$. Converting the sum into an integral via introducing the DOS, (6.94)

$$\begin{aligned} \frac{1}{N(\epsilon_F)V} &= \int_0^{\beta_c \hbar \omega_D/2} \frac{\tanh x}{x} dx, \quad \text{where, } x = \frac{\beta E_k}{2} \\ \int_0^{\beta_c \hbar \omega_D/2} \frac{\tanh x}{x} dx &= \ln\left(\frac{2e^\gamma}{\pi} \beta_c \hbar \omega_D\right) \end{aligned} \quad (6.94)$$

where γ denotes the Euler's constant, which has a value, $\gamma = 0.577\dots$ and thus $\frac{2e^\gamma}{\pi} \simeq 1.14$. This yields an expression for the transition temperature, T_c above which superconductivity vanishes,

$$k_B T_c = 1.13 \hbar \omega_c e^{-1/N(\epsilon_F)V}. \quad (6.95)$$

One may recall that earlier we have obtained an expression for the energy gap, namely,)

$$\Delta = 2 \hbar \omega_D e^{-1/N(\epsilon_F)V}$$

which was subsequently equated to $k_B T_c$. Also, a relationship between the energy gap at zero temperature to that of T_c is obtained as,

$$\frac{\Delta(0)}{k_B T_c} = \frac{2}{1.13} = 1.764. \quad (6.96)$$

Thus the gap at $T = 0$ is indeed comparable to the magnitude of $k_B T_c$. The numerical) factor has been tested in many experiments and found to be reasonable. The value of $2\Delta(0)$ varies between $3k_B T_c$ to $4.5k_B T_c$ for the conventional (that is, BCS) superconductors, and is mostly clustered around the BCS value $3.5k_B T_c$.

Finally, to arrive at an explicit temperature dependence of the gap function, we consider the integral form of the gap equation, namely,

$$\frac{1}{N(\epsilon_F)V} = \int_0^{\hbar \omega_D} \frac{\tanh \left[\frac{1}{2} \beta (\xi^2 + \Delta^2)^{1/2} \right]}{(\xi^2 + \Delta^2)^{1/2}} d\xi. \quad (6.97)$$

Now $\Delta(T)$ needs to be computed numerically by self-consistently solving the above) equation. For weak coupling superconductors, for which $\hbar \omega_D \gg k_B T_c$, $\Delta(T)/\Delta(0)$ is a universal function of (T/T_c) which decreases monotonically from 1 at $T = 0$ to zero at $T = T_c$. With $\Delta(T)$ determined, the quasiparticle energies can be written as,

$$E_k = \sqrt{\xi_k^2 + \Delta^2(T)}.$$

In the vicinity of $T = 0$, the temperature variation is exponentially slow because of $e^{-\Delta/k_B T} \approx 0$ (see figure 6.18). Hence the tan hyperbolic term is insensitive to T and stays very close to 1. Physically speaking, Δ remains nearly a constant until a significant number of quasiparticles are thermally excited.

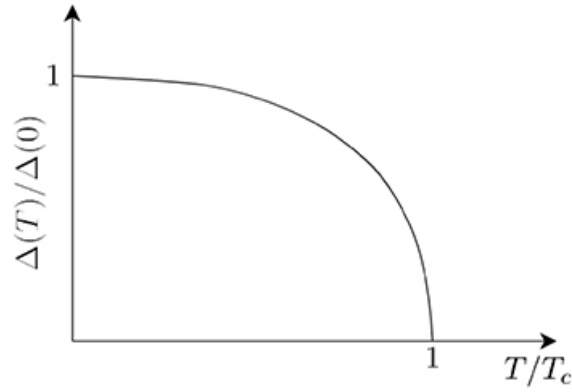


Figure 6.18. The scaled gap function $\Delta(T)/\Delta(0)$ is numerically obtained and plotted as a function of the reduced temperature, T/T_c . The area enclosed by the curve denotes the superconducting phase.

On the other hand, for $T \sim T_c$, $\Delta(T)$ vanishes with a vertical slope approximately as,

$$\frac{\Delta(T)}{\Delta(0)} \approx 1.74 \left[1 - \left(\frac{T}{T_c} \right) \right]^{1/2} \text{ for } T \rightarrow T_c \text{ from below.}$$

This square root dependence of the Δ on T is a hallmark feature of all mean field theories. For example, the magnetization vanishes in a similar fashion with temperature in molecular field theory of ferromagnetism.

6.4.2 Thermodynamics from BCS theory

The energy of a superconductor is obtained by explicitly solving the BCS Hamiltonian, which yields,

$$E = 2 \sum_{k > k_F} \xi_k v_k^2 + 2 \sum_{k < k_F} |\xi_k| u_k^2 + \sum_{kk'} V_{kk'} u_k v_k u_{k'} v_{k'} \quad (6.98)$$

where,

$$v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \right)$$

with $\xi_k = \varepsilon_k - \mu$, and $\Delta_k = -\sum_{k'} V_{kk'} u_k v_{k'}$. Further, the form of the attractive interaction is,

$$V_{kk'} = \begin{cases} -V & \text{for both } |\xi_k| \text{ and } |\xi_{k'}| < \hbar\omega_D \\ 0 & \text{otherwise.} \end{cases}$$

Also, we have obtained that,

$$\Delta \simeq 2\hbar\omega_D \exp[-1/N(\varepsilon_F)V].$$

Thus the energy difference between the superconducting and the normal states is given by,

$$E_s - E_n = -\frac{1}{2}N(\varepsilon_F)\Delta^2 \left[1 - \exp\left(-\frac{2}{N(\varepsilon_F)V}\right) \right] \quad (6.99)$$

which in the weak coupling limit assumes a form,

$$E_s - E_n = -\frac{1}{2}N(\varepsilon_F)\Delta^2$$

To enumerate the jump in the specific heat from BCS theory, we consider the entropy for the SC state given by [12],

$$S_s = -2k_B \sum_k [(1 - f_k) \ln(1 - f_k) + f_k \ln f_k]. \quad (6.10)$$

Further, in terms of the probability of a microstate being occupied, the entropy has a form,

$$S = -k_B \sum p_i \ln p_i. \quad (6.10)$$

Following this, the specific heat can be determined by,

$$C_{es} = T \frac{dS_s}{dT} = T \frac{dS_s}{d\beta} \cdot \frac{d\beta}{dT} = T \frac{dS_s}{d\beta} \cdot \left(-\frac{1}{k_B T^2} \right) = -\beta \frac{dS_s}{d\beta}$$

now,

$$\begin{aligned} C_s &= 2\beta k_B \sum_k \frac{\partial f_k}{\partial \beta} \ln \frac{f_k}{1 - f_k} \\ &= -2\beta^2 k_B \sum_k E_k \frac{\partial f_k}{\partial \beta} = -2\beta^2 k_B \sum_k E_k \frac{df_k}{d(\beta E_k)} \left(E_k + \beta \frac{dE_k}{d\beta} \right) \\ &= -2\beta k_B \sum_k \left(\frac{df_k}{dE_k} \right) \left(E_k^2 + \frac{1}{2}\beta \frac{d\Delta^2}{d\beta} \right). \end{aligned} \quad (6.10)$$

The first term is the contribution of the quasiparticles to the specific heat, while the second term is due to the temperature dependent gap function. The features of this specific heat are as follows:

(i) C_{es} is exponentially small at $T \ll T_c$, where the excitation energy Δ is much greater than $k_B T$. This explains the exponential dependence at low temperatures, namely $C_s \sim T_c e^{-T/T_c}$.

(ii) Near T_c , the gap, $\Delta(T) \rightarrow 0$, when one can replace E_k by $|\xi_k|$, the first term reduces to

$$C_s(T) = \gamma T = \frac{2\pi^2}{3} N(\varepsilon_F) k_B^2 T$$

which is continuous at $T = T_c$.

(iii) The second term is finite below T_c , where $\frac{d\Delta^2}{dT}$ is large but is zero above T_c , thereby giving rise to a discontinuity, ΔC in the electronic specific heat at T_c . This discontinuity is characteristic of a second-order phase transition.

The jump in the specific heat is enumerated as,

$$\begin{aligned} \Delta C &= (C_s - C_n) |_{T=T_c} \\ &= N(\varepsilon_F) k_B \beta^2 \frac{d(\Delta^2)}{d\beta} \int_{-\infty}^{\infty} \left(\frac{df}{d|\xi|} \right) d\xi \\ &= N(\varepsilon_F) \left(-\frac{d\Delta^2}{dT} \right)_{T=T_c}. \end{aligned} \tag{6.10}$$

Here, $\frac{df}{d|\xi|} = \frac{df}{d\xi}$ as f is an even function of ξ . Further, using⁴

$$\Delta(T) = 1.74\Delta(0) \left(1 - \frac{T}{T_c}\right)^{1/2}$$

the magnitude ΔC can be computed to yield,

$$\Delta C = 9.4 N(\varepsilon_F) k_B^2 T_c$$

where $\Delta(0) = 1.76 k_B T_c$. The normalized discontinuity, usually measured in experiments is given by,

$$\frac{\Delta C}{C_{\text{en}}} = \frac{9.4}{2\pi^2/3} \approx 1.43$$

$C_{\text{es}}(T)$ can be found numerically from the parent expression (equation (6.102)). $C_{\text{es}}(T)$ can be integrated to obtain $U_{\text{es}}(T)$. The clue is that at $T = T_c$, $U_{\text{es}}(T)$ must be same as the normal value $U_{\text{en}}(0) + \frac{1}{2} \gamma T_c^2$.

6.5 Electromagnetic considerations

Let us consider a system of N electrons subjected to a magnetic field, $\mathbf{B}$ which is described by a vector potential. The momentum for each of these particles described by, $i = 1, \dots, N$ under an external vector potential, $\mathbf{A}(\mathbf{r}, t)$ can be replaced by,

$$\mathbf{p}_i \rightarrow \mathbf{p}_i + e\mathbf{A}(\mathbf{r}).$$

This yields a Hamiltonian,

$$\mathcal{H} = \frac{1}{2m} \sum_{i=1}^N (\mathbf{p}_i - e\mathbf{A}(\mathbf{r}_i))^2 + \hat{V} = \mathcal{H}_0 + \mathcal{H}' \quad (6.10)$$

where, $\mathcal{H}_0$ is the many-body Hamiltonian with $\mathcal{H}_0 = \sum_i \frac{p_i^2}{2m}$. To the linear order in $\mathbf{A}$,⁴⁾ the coupling to the external probe can be written as,

$$\mathcal{H}' = -\frac{e}{2m} \sum_{i=1}^N [\mathbf{p}_i \cdot \mathbf{A}(\mathbf{r}_i) + \mathbf{A}(\mathbf{r}_i) \cdot \mathbf{p}_i]. \quad (6.10)$$

With the choice of a transverse gauge, that is, $\nabla \cdot \mathbf{A} = 0$, both the terms yield the same⁵⁾ contribution. Hence we can write,

$$\mathcal{H}' = -\frac{e}{m} \sum_{i=1}^N \mathbf{p}_i \cdot \mathbf{A}(\mathbf{r}_i). \quad (6.10)$$

This greatly simplifies the analysis, as will be clear in the following discussion.⁶⁾

The total current operator can be written as,

$$\mathbf{j}(\mathbf{r}) = \frac{e}{2} \sum_{i=1}^N (\mathbf{v}_i \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \mathbf{v}_i) \quad (6.10)$$

with the velocity operator given by,⁷⁾

$$\mathbf{v}_i = \frac{1}{m} (\mathbf{p}_i - \frac{e}{c} \mathbf{A}(\mathbf{r}_i)) \quad (6.10)$$

and the δ -functions denote the positions of the particles. The above form for current can⁸⁾ be split into,

$$\mathbf{j}(\mathbf{r}) = \mathbf{j}_p(\mathbf{r}) + \mathbf{j}_d(\mathbf{r}) \quad (6.10)$$

where $\mathbf{j}_p$ and $\mathbf{j}_d$ denote the paramagnetic and diamagnetic current densities, and are given⁹⁾ by the following.

$$\mathbf{j}_p(\mathbf{r}) = \frac{e}{2m} \sum_{i=1}^N (\mathbf{p}_i \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \mathbf{p}_i).$$

In terms of the fermionic operators, (6.11
0)

$$\mathbf{j}_p(\mathbf{q}) = \frac{e\hbar}{m} \sum_{\mathbf{k}, \sigma} (\mathbf{k} + \mathbf{q}/2) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}+\mathbf{q}, \sigma} \quad (6.11)$$

where $\mathbf{k}$ and $\mathbf{q}$ denote momentum indices. Next, the diamagnetic current density is written as, 1)

$$\mathbf{j}_d(\mathbf{r}) = -\frac{e^2}{m} \mathbf{A}(\mathbf{r}) \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (6.11)$$

which again in terms of the fermionic operators can be expressed via, 2)

$$\mathbf{j}_d(\mathbf{q}) = -\frac{e^2}{m} \mathbf{A}_\mathbf{q} \sum_{\mathbf{k}, \sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} = -\frac{ne^2}{m} \mathbf{A}_\mathbf{q}. \quad (6.11)$$

Using these definitions, the coupling to the vector potential is expressed as, 3)

$$\mathcal{H}_I = -\frac{1}{\Omega} \int d\mathbf{r} \mathbf{j}_p(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}) = -\frac{1}{\Omega} \sum_{\mathbf{q}} \mathbf{j}_p(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q}) \quad (6.11)$$

where Ω denotes the volume and can be set to unity for convenience. It may be noted that the external probe $\mathbf{A}$ couples only to the paramagnetic current $\mathbf{j}_p$ in the Hamiltonian, $\mathcal{H}_I$. $\mathbf{j}_d$ is already linear in $\mathbf{A}$. So it is only needed to evaluate $\mathbf{j}_p$ at the linear response level, and hence add to $\mathbf{j}_d$ to get the total current response, such that, 4)

$$\langle \mathbf{j} \rangle = \langle \mathbf{j}_p \rangle + \langle \mathbf{j}_d \rangle. \quad (6.11)$$

According to linear response theory one can write, 5)

$$\langle j_\alpha \rangle(\mathbf{q}, \omega) = \chi_{\alpha\beta}(\mathbf{q}, \omega) A_\beta(\mathbf{q}, \omega) - \frac{e^2}{m} \left\langle \sum_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} \right\rangle A_\alpha(\mathbf{q}, \omega) \quad (6.11)$$

where ω denotes the frequency and the current–current response function is given by, 6)

$$\chi_{\alpha\beta} \equiv \chi_{j_\alpha j_\beta}^p(\mathbf{q}, \omega)$$

with the density appearing in the second term,

$$n \equiv \left\langle \sum_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} \right\rangle.$$

For obtaining the Meissner effect, we are interested in the zero momentum (long wavelength) and zero frequency (static) response of $\chi_{\alpha\beta}$, that is, for $\omega = 0, q \rightarrow 0$. One finally gets for the

current density,

$$\begin{aligned}\langle j_\alpha \rangle(\mathbf{q} \rightarrow 0, \omega = 0) &= \chi_{\alpha\beta}(\mathbf{q} \rightarrow 0, \omega = 0) A_\beta(\mathbf{q} \rightarrow 0, \omega = 0) \\ &- \frac{ne^2}{m} A_\alpha(\mathbf{q} \rightarrow 0, \omega = 0).\end{aligned}$$

For computing conductivity, we have to consider the finite frequency limit ($\omega \neq 0$)⁵. Consider the electric field (the scalar potential is set to zero because of the gauge condition),

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = i\omega \mathbf{A}. \quad (6.11 \quad 7)$$

Thus,

$$\langle j_\alpha \rangle = \sigma_{\alpha\beta} E_\beta = i\omega \sigma_{\alpha\beta} A_\beta \quad (6.11 \quad 8)$$

which subsequently yields,

$$\sigma_{\alpha\beta} = \frac{1}{i\omega} \left[\chi_{\alpha\beta}(q = 0, \omega) - \frac{ne^2}{m} \delta_{\alpha\beta} \right] \quad (6.11 \quad 9)$$

here,

$$\sigma_{\alpha\beta} = \Re \sigma_{\alpha\beta} + \Im \sigma_{\alpha\beta}. \quad (6.12 \quad 0)$$

Finally at the linear response level, the Kubo formula for the conductivity tensor is written as,

$$\sigma_{\alpha\beta}(\omega) = \frac{1}{i\omega} \left[\chi_{j_p \alpha j_p \beta}(q = 0, \omega) - \frac{ne^2}{m} \delta_{\alpha\beta} \right]. \quad (6.12 \quad 1)$$

6.5.1 Meissner effect

We demonstrate the expulsion of the magnetic field or the Meissner effect in the following.

$$\mathbf{j}_p = \frac{e\hbar}{m} \sum_{\mathbf{k}} (\mathbf{k} + \mathbf{q}/2) (c_{\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}+\mathbf{q}\uparrow} + c_{-\mathbf{k}\downarrow}^\dagger c_{-(\mathbf{k}+\mathbf{q})\downarrow}). \quad (6.12 \quad 2)$$

The creation and the annihilation operators are written in terms of the quasiparticle operators via,

$$\begin{aligned}c_{\mathbf{k}\uparrow} &= u_{\mathbf{k}} \gamma_{\mathbf{k}\uparrow} + v_{\mathbf{k}} \gamma_{-\mathbf{k}\downarrow}^\dagger \\ c_{-\mathbf{k}\downarrow}^\dagger &= -v_{\mathbf{k}} \gamma_{\mathbf{k}\uparrow} + u_{\mathbf{k}} \gamma_{-\mathbf{k}\downarrow}\end{aligned} \quad (6.12 \quad 3)$$

where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are the coherence factors. These relationships are canonical, since they preserve the fermionic anticommutation relations. This allows the paramagnetic current to be

written as,

$$\begin{aligned}
 j_p = \frac{e\hbar}{m} \sum_k (\mathbf{k} + \mathbf{q}/2) & [(uu' + vv')(\gamma_{k\uparrow}^\dagger \gamma_{k+q\uparrow} - \gamma_{-(k+q)\downarrow}^\dagger \gamma_{-k\downarrow}) \\
 & + (uv' - vu')(\gamma_{k\uparrow}^\dagger \gamma_{-(k+q)\downarrow}^\dagger - \gamma_{-k\downarrow}^\dagger \gamma_{k+q\uparrow})]
 \end{aligned} \tag{6.12}$$

where the momentum indices are dropped by redefining,

$$u = u_k \quad v = v_k \quad u' = u_{k+q} \quad v' = v_{k+q}.$$

Now in the linear regime, the current–current response function is written as,

$$\chi_{\alpha\beta}(q, \omega = 0) = \sum_{m,n} \frac{e^{-\beta\omega_m}}{Z} \left\{ \frac{(j_p^\alpha(q))_{m,n} (j_p^\beta(-q))_{n,m}}{\omega_{nm}} + \text{c. c.} \right\}. \tag{6.12}$$

One can import the contribution of j_p from equation (6.124). In particular we look at different terms below.

(i) $\gamma^\dagger \gamma$ term with $\sigma = \uparrow$

$$\sum_{k, k'} f(u, v) (k' + q/2)^\alpha (k - q/2)^\beta \langle m | \gamma_{k'}^\dagger \gamma_{k'+q} | n \rangle \langle n | \gamma_k^\dagger \gamma_{k-q} | m \rangle \frac{1}{\omega_{nm}}$$

where $f(u, v)$ is the coherence factor determined later. The only $|n\rangle$ that contributes is given by $|n\rangle = \gamma_k^\dagger \gamma_{k-q} |m\rangle$. Thus,

$$\mathbf{k}' + \mathbf{q} = \mathbf{k}, \quad \text{and} \quad , \quad \omega_{nm} \equiv \omega_n - \omega_m = E_k - E_{k-q}.$$

This yields the above contribution to have the form.

$$\begin{aligned}
 \sum_k f(u, v) \underbrace{(k - q/2)^\alpha (k - q/2)^\beta}_{E_k - E_{k-q}} &= \sum_m \frac{e^{-\beta\omega_m}}{Z} \langle m | \gamma_{k-q}^\dagger \gamma_k \gamma_k^\dagger \gamma_{k-q} | m \rangle \\
 &= \langle \gamma_{k-q}^\dagger \gamma_{k-q} (1 - \gamma_k^\dagger \gamma_k) \rangle \\
 &= f_{k-q} (1 - f_k).
 \end{aligned} \tag{6.12}$$

The above being the thermal average, we get the Fermi distribution function, f_k in the above expression. Finally, the coherence factor, $f(u, v)$ is,

$$\begin{aligned}
 &= (u_{k'} u_{k'+q} + v_{k'} v_{k'+q}) (u_k u_{k-q} + v_k v_{k-q}) \\
 &= (u_k u_{k-q} + v_k v_{k-q})^2, \quad \text{since} \quad \mathbf{k}' = \mathbf{k} - \mathbf{q}.
 \end{aligned}$$

One may shift the dummy index, $\mathbf{k} \rightarrow \mathbf{k} + \mathbf{q}$ which makes the complex conjugate equal to the term calculated above. This yields the term with $\gamma^\dagger \gamma$ for $\uparrow$ -spin as,

$$= \frac{2e^2 \hbar^2}{m^2} \sum_k \frac{(uu' + vv')^2 f(1 - f')}{E' - E} (k + q/2)^\alpha (k + q/2)^\beta.$$

(ii) *Combination of $\gamma^\dagger \gamma$ with $\sigma = \downarrow$*

Same as (i) except for a negative sign in front of $\gamma_\downarrow^\dagger \gamma_\downarrow$ comes in $(-1)^2$ in χ and $\mathbf{k} \rightarrow \mathbf{k} + \mathbf{q}$ in u' s, v' s, f' s and E' s. This yields for the $\gamma^\dagger \gamma$ term corresponding to $\downarrow$ -spin,

$$= \frac{2e^2 \hbar^2}{m^2} \sum_k \frac{(uu' + vv')^2 f'(1 - f)}{E - E'} (k + q/2)^\alpha (k + q/2)^\beta.$$

Subtracting (ii) from (i), that is, contributions from spin- $\uparrow$ and spin- $\downarrow$,

$$= \frac{2e^2 \hbar^2}{m^2} \sum_k (k + q/2)^\alpha (k + q/2)^\beta (uu' + vv')^2 \frac{(f - f')}{E - E'}.$$

(iii) and (iv) *The combination of the $\gamma^\dagger \gamma^\dagger$ and $\gamma \gamma$ terms*

This comes with a coherence factor $(u_k v_{k+q} - v_k u_{k+q}) \rightarrow 0$ as $q \rightarrow 0$ so they will not contribute, however, we shall retain them for considering the finite q response. They can be written as,

$$\begin{aligned} & \sum_{m,n} \frac{e^{-\beta \omega_m}}{Z} \langle m | \gamma_k^\dagger \gamma_{-(k+q)}^\dagger | n \rangle \langle n | \gamma_{-(k+q)} \gamma_k | m \rangle \frac{(-1)}{\omega_{nm}} \\ &= \frac{(-1) \langle \gamma_k^\dagger \gamma_k \gamma_{-(k+q)}^\dagger \gamma_{-(k+q)} \rangle}{-E_k - E_{k+q}} = \frac{f f'}{E + E'}. \end{aligned}$$

The negative sign, that is, -1 is the only difference between the $\gamma^\dagger \gamma$ and the $\gamma \gamma$ terms The other one is,

$$= \frac{(-1) \langle \gamma_{-(k+q)} \gamma_k \gamma_k^\dagger \gamma_{-(k+q)}^\dagger \rangle}{E_k + E_{k+q}} = \frac{(-1)(1 - f)(1 - f')}{E + E'}.$$

Here the coherence factor is written as,

$$(uv' - vu')(u'v - v'u) = -(uv' - vu')^2.$$

Adding contributions of (iii) and (iv)

$$= \frac{(1 - f + f')}{E + E'} (uv' - vu')^2 \times 2.$$

The current–current response function is written as,

$$\chi_{\alpha\beta}(\mathbf{q}, \omega = 0) = \frac{2e^2\hbar^2}{m^2} \sum_{\mathbf{k}} (k + q/2)^\alpha (k + q/2)^\beta \times \left[(uv' + vu')^2 \frac{(f - f')}{E - E'} + (uv' - vu')^2 \frac{(1 - f - f')}{E + E'} \right]. \quad (6.12)$$

Now consider special cases of this general formula, that is, $\mathbf{q} \rightarrow 0$ response which yields, 7)

$$(uv' - vu') \rightarrow 0 \quad (uv' + vu') \rightarrow 1.$$

In which case,

$$\chi_{\alpha\beta}(\mathbf{q} \rightarrow 0, \omega = 0) = \frac{2e^2\hbar^2}{m^2} \sum_{\mathbf{k}} k_\alpha k_\beta \left(-\frac{\partial f_k}{\partial E_k} \right). \quad (6.12)$$

The same result can be obtained in a more direct manner [6]. 8)

In the large wavelength limit, namely, $\mathbf{q} = 0$

$$\mathbf{j}_p = \frac{e\hbar}{m} \sum_{\mathbf{k}} \mathbf{k} \left(\gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} - \gamma_{-k\downarrow}^\dagger \gamma_{-k\downarrow} \right)$$

$$\langle \mathbf{j}_p \rangle = \frac{e\hbar}{m} \sum_{\mathbf{k}} \mathbf{k} (f_{k\uparrow} - f_{-k\downarrow}). \quad (6.12)$$

We must keep track of how the quasiparticle spectrum, $E_{k\uparrow}$, changes under the influence of the external Hamiltonian, 9)

$$\mathcal{H}' = -\frac{1}{c} \sum_{\mathbf{q}} \mathbf{j}_1(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q})$$

$$= -\frac{e\hbar}{mc} \sum_{\mathbf{k}} (\mathbf{k} \cdot \mathbf{A}_0) (\gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} - \gamma_{-k\downarrow}^\dagger \gamma_{-k\downarrow}) \quad (6.13)$$

where the $q = 0$ term is retained in the last step of the equation. This yields the energy spectrum that is reminiscent of spin splitting in a magnetic field. The energies are modified as, 0)

$$E_{k\uparrow} \rightarrow E_{k\uparrow} - \frac{e\hbar}{mc} \mathbf{k} \cdot \mathbf{A}_0$$

Similarly,

$$E_{-k\downarrow} \rightarrow E_{-k\downarrow} + \frac{e\hbar}{mc} \mathbf{k} \cdot \mathbf{A}_0$$

where $\mathbf{A}_0$ is the vector potential for $q = 0$. Now,

$$\begin{aligned} f_{k\uparrow} - f_{-k\downarrow} &\approx \left(-\frac{\partial f_k}{\partial E_k} \right) \frac{2e\hbar}{mc} \mathbf{k} \cdot \mathbf{A}_0 \\ \langle \mathbf{j}_p(\mathbf{q} = 0) \rangle &= \frac{2e^2\hbar^2}{m^2c} \sum_{\mathbf{k}} (\mathbf{k} \cdot \mathbf{A}_0) \mathbf{k} \left(-\frac{\partial f_k}{\partial E_k} \right). \end{aligned} \quad (6.13)$$

Since, by symmetry $\langle \mathbf{j}_p(\mathbf{q} = 0) \rangle$ is parallel to $\mathbf{A}_0$. Also,

$$\int \frac{d\Omega}{4\pi} (\mathbf{k} \cdot \hat{A}_0)^2 = \frac{k^2}{3}, \quad \text{as } \langle \cos^2 \theta \rangle = 1/3.$$

Therefore, $\hat{A}_0$ being the unit vector along A_0 ,

$$\begin{aligned} \langle \mathbf{j}_p(0) \rangle &= \frac{2}{3} \frac{e^2\hbar^2}{m^2c} \sum_{\mathbf{k}} k^2 \left(-\frac{\partial f}{\partial E_k} \right) \mathbf{A}_0 \\ \text{Since, } \sum_{\mathbf{k}} k^2 \left(-\frac{\partial f}{\partial E_k} \right) &\approx k_F^2 N(0) \int d\xi_k \left(-\frac{\partial f}{\partial E_k} \right) \end{aligned} \quad (6.13)$$

where $N(0)$ is the density of states at the Fermi level. Moreover, the pre-factor of $\int d\xi_k$ is given by,

$$= \frac{2}{3} \frac{e^2\hbar^2}{m^2} k_F^2 \frac{3n}{4\varepsilon_F} = \frac{ne^2}{m}.$$

Hence, the average paramagnetic current is given by,

$$\langle \mathbf{j}_p(q = 0) \rangle = \frac{ne^2}{m} \int d\xi_k \left(-\frac{\partial f}{\partial E_k} \right) \mathbf{A}_0. \quad (6.13)$$

The normal state can be obtained by putting $\Delta = 0$ in the expression for quasiparticle energies, that are given by, $E = \sqrt{\xi^2 + \Delta^2}$. This yields,

$$E = |\xi| \quad \text{Also,} \quad \int d\xi \left(-\frac{\partial f}{\partial \xi} \right) = 1.$$

Finally, one gets,

$$\langle \mathbf{j}_p(q = 0) \rangle = \frac{ne^2}{m} \mathbf{A}_0. \quad (6.13)$$

4)

Hence, the paramagnetic response, $\langle \mathbf{j}_p \rangle$ and the diamagnetic response $\langle \mathbf{j}_d \rangle$ exactly cancel each other, thereby yielding the total induced current, $\langle \mathbf{j} \rangle = 0$. Consequently, no circulating currents are induced in a normal metal in equilibrium condition and thus the external magnetic field $\mathbf{H}$ is not shielded.

One can ignore weak Landau diamagnetic effects, which arise from the fact that,

$$\langle \mathbf{j}_p(\mathbf{q} \rightarrow 0) \rangle = \left(\frac{ne^2}{m} - \alpha q^2 \right) \mathbf{A}(q) \quad (6.135)$$

where one can write,

$$\mathbf{A}_{\text{total}} = \left(\frac{ne^2}{m} - \alpha q^2 \right) \mathbf{A}(q).$$

The total current density now becomes,

$$\langle \mathbf{j}(\mathbf{q} \rightarrow 0) \rangle = -\alpha q^2 \mathbf{A}_{\text{total}} \quad (\alpha > 0) \quad (6.136)$$

where α is a positive quantity, and $\mathbf{A}_{\text{ind}}$ obeys the wave equation,

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \mathbf{A}_{\text{ind}} = \mu_0 \langle \mathbf{j} \rangle. \quad (6.137)$$

In a static situation, the contribution of the term $\frac{1}{c^2} \frac{\partial^2}{\partial t^2} = 0$. Then,

$$q^2 \mathbf{A}_{\text{ind}} = -\mu_0 \times q^2 \mathbf{A}_{\text{total}}. \quad (6.138)$$

Using self-consistent fields,

$$\begin{aligned} \mathbf{A}_{\text{total}} &= \mathbf{A}_{\text{ind}} + \mathbf{A}_{\text{ext}} \\ \text{or, } \mathbf{A}_{\text{total}} &= \frac{\mathbf{A}_{\text{ext}}(\mathbf{q})}{1 + \mu_0 \alpha} \quad (\alpha > 0). \end{aligned} \quad (6.139)$$

Hence the total vector potential derives contribution from the external field and $\mathbf{A}_{\text{ind}}$.

Taking curl to obtain the magnetic field,

$$|\mathbf{H}_{\text{total}}| = \frac{|\mathbf{H}_{\text{ext}}|}{1 + \eta} \quad (6.140)$$

where η is a positive quantity, such that $\mathbf{H}_{\text{total}} < \mathbf{H}_{\text{ext}}$ which means that the magnetic field is somewhat screened, and the system shows weak diamagnetism.

Let us now discuss the superconducting state which is characterized by $\Delta \neq 0$ and the cancellation of the paramagnetic response $\langle \mathbf{j}_p \rangle$ and the diamagnetic response $\langle \mathbf{j}_d \rangle$ is not valid

any longer. In fact, as we shall see, at $T = 0$, $\langle \mathbf{j}_p \rangle = 0$ and therefore only the diamagnetic response $\langle \mathbf{j}_d \rangle$ contributes.

To this effect, let us define the ratio of the density of normal electrons to that of the total density, namely, $\frac{n_n}{n}$. This quantity will vanish in the superconducting state, or as, $T \rightarrow T_c^-$, that is, when T approaches T_c from below.

$$\begin{aligned} \frac{n_n}{n} &= \int_{-\infty}^{\infty} d\xi_k \left(-\frac{\partial f_k}{\partial \xi} \right) = 2 \int_{\Delta}^{\infty} dE \frac{E}{\sqrt{E^2 - \Delta^2}} \left(-\frac{\partial f}{\partial E} \right) \\ &= \frac{1}{2T} \int_0^{\infty} d\xi \operatorname{sech}^2 \left(\frac{\sqrt{\xi^2 + \Delta^2(T)}}{2T} \right) = Y(T) \end{aligned} \quad (6.14)$$

where $Y(T)$ is known as the Yosida function. The Yosida function characterizes the ¹⁾ response of quantities to which other pairs do not contribute, and thus represents the response of the normal fluid. Let us explore the following limits, namely, $T \ll T_c$ and $T \rightarrow T_c^-$.

- (i) For $T \ll T_c$, $Y(T)$ behaves as, $Y(T) \sim e^{-\Delta(0)/T}$.
- (ii) For $T \rightarrow T_c^-$, $Y(T) \approx 1 - 2 \frac{(T_c - T)}{T_c}$.

The $q = 0$ components of the paramagnetic and the diamagnetic contributions are respectively written as,

$$\begin{aligned} \langle \mathbf{j}_p \rangle_{q=0} &= \frac{ne^2}{m} \left(\frac{n_n}{n} \right) \mathbf{A}_0 \\ \langle \mathbf{j}_d \rangle_{q=0} &= -\frac{ne^2}{m} \mathbf{A}_0. \end{aligned} \quad (6.14)$$

Thus,

$$\langle \mathbf{j} \rangle = \langle \mathbf{j}_p + \mathbf{j}_d \rangle = -\frac{ne^2}{m} \left(1 - \frac{n_n}{n} \right) \mathbf{A}_0 \equiv -\frac{n_s e^2}{m} \mathbf{A}_0 \quad (6.14)$$

where one can define the superelectron density, that is, $\frac{n_s}{n}$ ³⁾

$$\frac{\rho_s}{\rho} = 1 - \frac{\rho_n}{\rho}.$$

So,

$$\langle \mathbf{j} \rangle = -\frac{n_s e^2}{m} \mathbf{A}. \quad (6.14)$$

Thus the superelectron density can be described in terms of the Yoshida function as, ⁴⁾

$$\frac{n_s(T)}{n} = 1 - Y(T).$$

Hence the London penetration depth (or the inverse of it) is expressed as,

$$\frac{1}{\lambda_L^2(T)} = \frac{4\pi n_s(T)e^2}{m} = \frac{1}{\lambda_L^2(0)} (1 - Y(T)). \quad (6.14)$$

Thus the temperature dependence of the penetration depth is expressed by the temperature variation of the Yoshida function, $Y(T)$ (see figure 6.19).

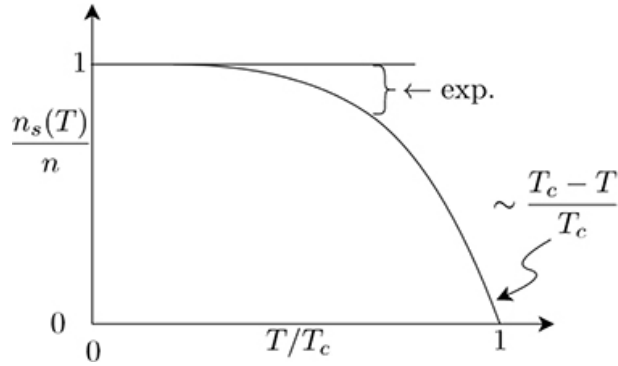


Figure 6.19. The behaviour of the superfluid density scaled by the total density, $\frac{n_s(T)}{n}$ is shown as a function of reduced temperature, T/T_c . The behaviour of the superfluid density close to $T = T_c$ is given by $(1 - \frac{T}{T_c})$.

6.5.2 Electromagnetic response in the transverse gauge

An externally applied vector potential, $\mathbf{A}_{\text{ext}}(\mathbf{r}, t)$, induces current in the system $\langle \mathbf{j}(\mathbf{r}, t) \rangle$, which in turn generates an electromagnetic field. Using Maxwell's equation we have,

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \mathbf{A}_{\text{ind}}(\mathbf{r}, t) = \mu_0 \langle \mathbf{j}(\mathbf{r}, t) \rangle \quad (6.14)$$

which follows from,

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} \quad \text{and} \quad \nabla \times (\nabla \times \mathbf{A}) = -\nabla^2 \mathbf{A}$$

in a gauge where $\nabla \cdot \mathbf{A} = 0$. Therefore, Fourier transforming one obtains,

$$\left(q^2 - \frac{\omega^2}{c^2} \right) \mathbf{A}_{\text{ind}}(\mathbf{q}, \omega) = \mu_0 \langle \mathbf{j}(\mathbf{q}, \omega) \rangle. \quad (6.14)$$

The total vector potential is written as,

7)

$$\mathbf{A}_{\text{tot}} = \mathbf{A}_{\text{ext}} + \mathbf{A}_{\text{ind}}$$

and for the current density,

$$\begin{aligned} \langle \mathbf{j}(\mathbf{q}, \omega) \rangle &= \left(\chi^T - \frac{ne^2}{m} \right) \mathbf{A}_{\text{tot}} = -\frac{n_s e^2}{m} \mathbf{A}_{\text{ext}} \\ &= -\frac{1}{\lambda_L^2} \mathbf{A}_{\text{tot}}. \end{aligned} \tag{6.14}$$

Now,

$$\begin{aligned} \mathbf{B} &= \mu_0 (\mathbf{H} + 4\pi \mathbf{M}) \\ \text{also } \mathbf{B} &= \nabla \times \mathbf{A}_{\text{tot}} \\ \text{and } \mathbf{H} &= \nabla \times \mathbf{A}_{\text{ext}} \quad \text{that is, } \mathbf{M} = \frac{1}{\mu_0} \nabla \times \mathbf{A}_{\text{ind}}. \end{aligned} \tag{6.14}$$

In the static limit, that is, $\omega = 0$,

$$\begin{aligned} \mathbf{A}_{\text{ind}} &= \frac{4\pi}{c} \frac{1}{q^2} \langle \mathbf{j}(\mathbf{q}, \omega) \rangle \\ \text{or, } \mathbf{A}_{\text{ind}} &= \frac{-1/\lambda_L^2(T)}{q^2 + 1/\lambda_L^2(T)} \mathbf{A}_{\text{ext}}. \end{aligned} \tag{6.15}$$

Thus the susceptibility can be written as,

$$\chi = \frac{\partial M}{\partial H} = \frac{\frac{1}{\lambda_L^2(T)}}{q^2 + \frac{1}{\lambda_L^2(T)}} \quad \text{that is, } \lim_{q \rightarrow 0} \chi = -1. \tag{6.15}$$

The above equation implies the condition of perfect diamagnetism. The superconductors are known to be perfect diamagnets. It may be noted that the diamagnetic susceptibility of metals is usually of the order of 10^{-5} – 10^{-6} .

6.6 Ginzburg–Landau (GL) theory

In 1937 Ginzburg and Landau [13] developed a model to describe second-order phase transition⁶ that is, those transitions which involve latent heat. It is physically meaningful to state that the quantum many-body state, ψ (something similar to what we have seen in BCS theory) should be a function that minimizes the free energy, F_s of the superconducting state. Ginzburg and Landau considered F_s to be a functional of the wavefunction ψ , and hence use a variational principle to minimize F_s with respect to ψ , and obtain a set of differential equations for ψ that govern the physical properties of the superconductor in equilibrium conditions. To this end, a concept of the order parameter has been developed which smoothly vanishes at a

temperature $T > T_c$, while it is non-zero for $T < T_c$. The identification of the order parameter is often obvious from the properties of second-order phase transitions. For example, for a ferromagnetic to a paramagnetic transition, the order parameter is the magnetization, which is finite below the magnetic transition temperature, T_c and goes to zero above T_c , whereas, a superconducting state has an energy gap that separates the ground state and the first excited state, as the order parameter assumes non-zero values below the superconducting transition temperature T_c which, however, vanishes above T_c .

Since the order parameter evolves continuously from zero below T_c , it is natural to expand the free energy as a power series in the order parameter. The free energy is a scalar, but the order parameter, may in general be a vector, or even a complex quantity. We shall study GL theory by constructing the free energy for the superconductor in terms of the wavefunction for the ‘*superelectrons*’ as the order parameter (instead of the energy gap between the ground and the excited states), $\psi(\mathbf{r})$, the density of the superelectrons is $n_s = |\psi(\mathbf{r})|^2$.

Consider a homogeneous superconductor at zero external magnetic field, that is, $\mathbf{H} = 0$. Also, $\psi(\mathbf{r})$ is independent of $\mathbf{r}$. Let us expand the free energy near the transition from a normal metal to a superconductor, that is, near T_c ,

$$F_s = F_n + \alpha |\psi|^2 + \frac{\beta}{2} |\psi|^4 \quad (6.15)$$

where F_s denotes the free energy density of the superconductor, and F_n being the free energy density of the normal state. Further, α, β are the phenomenological expansion coefficients characterizing the material, and they depend on temperature. The minimum of F_s is obtained via minimizing F with respect to the order parameter,

$$\frac{dF_s}{d|\psi|^2} = 0. \quad (6.15)$$

The minimization implies that (writing the equilibrium value of ψ and ψ_0),

$$|\psi_0|^2 = -\alpha/2\beta. \quad (3)$$

Recall that $|\psi_0|^2$ is the density of the superelectrons that minimizes F_s . Now,

$$\begin{aligned} \frac{dF_s}{d\psi} &= 2\alpha\psi + 2\beta\psi^3 \\ &= 2\psi[\alpha + \beta\psi^2]. \end{aligned} \quad (6.15)$$

Thus, the difference in energies between the normal and the superconducting states is obtained from, (4)

$$\begin{aligned}
F_n - F_s &= \alpha \left(\frac{\alpha}{\beta} \right) - \frac{\beta}{2} \left(\frac{\alpha}{\beta} \right)^2 \\
&= \frac{\alpha^2}{\beta} - \frac{\alpha^2}{2\beta} = \frac{\alpha^2}{2\beta}.
\end{aligned}
\tag{6.155}$$

The temperature dependence of α and β can be deciphered as follows. At the first order in temperature, that is, when T is close to T_c , let us postulate a linear behaviour of the form,

$$\alpha \simeq a(T - T_c) \tag{6.156}$$

where a is independent of temperature. This implies,

$$\begin{aligned}
\alpha &= 0 \quad \text{at} \quad (T = T_c) \\
\alpha &< 0 \quad \text{at} \quad (T < T_c).
\end{aligned}
\tag{6.157}$$

Thus the order parameter assumes a form,

$$\begin{aligned}
|\psi| &= 0 \quad \text{for} \quad T > T_c \\
|\psi| &= \left[\frac{a(T - T_c)}{\beta} \right]^{1/2} \quad \text{for} \quad T < T_c
\end{aligned}
\tag{6.158}$$

α changes sign across the phase transition, and hence β should not change sign and remain a constant at least for small deviations in temperature from the transition point.

Next consider a superconductor in an external magnetic field, $\mathbf{H}_{\text{ext}}$. The electrons interact with the external field via their momenta transforming as,

$$\begin{aligned}
\mathbf{p} &\rightarrow \mathbf{p} - e\mathbf{A} \quad \text{where} \quad \mathbf{B}_{\text{ext}} = \nabla \times \mathbf{A} \\
\text{which implies,} \quad \nabla &\rightarrow \nabla - ie\mathbf{A}.
\end{aligned}
\tag{6.159}$$

Thus the free energies have to be written as,

$$F_s = F_n + \alpha |\psi|^2 + \frac{1}{2}\beta |\psi|^4 + \frac{\hbar^2}{2m} \left| \left[\nabla - \frac{ie}{\hbar c} A(\mathbf{r}) \right] \psi(\mathbf{r}) \right|^2 + \mu_0 H_{\text{ext}}^2(\mathbf{r}) \tag{6.160}$$

where $\mathbf{H}_{\text{ext}}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r})$. The free energy is a functional of

$$\{\psi(\mathbf{r}), \psi^*(\mathbf{r}), \nabla\psi(\mathbf{r}), \nabla\psi^*(\mathbf{r}), H(\mathbf{r})\}$$

and each one of these is a function of the spatial variable $\mathbf{r}$. Minimizing with respect to $\psi^*(\mathbf{r})$ yields,

$$\delta F_s = \left\{ -\frac{\hbar^2}{2m} \left(\nabla - \frac{ie}{\hbar} \mathbf{A}(\mathbf{r}) \right)^2 \psi(\mathbf{r}) + \alpha \psi(\mathbf{r}) + \beta |\psi(\mathbf{r})|^2 \psi(\mathbf{r}) \right\} \delta \psi^*(\mathbf{r}) \\ + \frac{\hbar^2}{2m} \left(\nabla - \frac{ie}{\hbar} \mathbf{A}(\mathbf{r}) \right) \psi(\mathbf{r}) \delta \psi^*(\mathbf{r}). \quad (6.16)$$

The last term can be written as, $\mu_0 H^2$, or $\mu_0 (\nabla \times \mathbf{A})^2$. Further, variation with respect to ψ yields the complex conjugate of this equation. Putting $\delta F_s = 0$ for an arbitrary variation of the order parameter, namely, $\delta \psi^*(\mathbf{r})$, we get the first GL equation,

$$-\frac{\hbar^2}{2m} \left(\nabla - \frac{ie}{\hbar} \mathbf{A}(\mathbf{r}) \right)^2 \psi(\mathbf{r}) + \alpha \psi(\mathbf{r}) + \beta |\psi(\mathbf{r})|^2 \psi(\mathbf{r}) = 0 \quad (6.16)$$

where we have neglected the magnetic energy density due to the external field, that is, the last term. Now minimizing it with respect to $\mathbf{A}(\mathbf{r})$ yields Ampere's law,

$$\nabla \times \mathbf{H}(\mathbf{r}) = \mu_0 \mathbf{j}(\mathbf{r}) \quad (6.16)$$

where,

$$\mathbf{j}(\mathbf{r}) = -\frac{ie\hbar}{2m} \left[\psi^*(\mathbf{r}) \nabla \psi(\mathbf{r}) - \psi(\mathbf{r}) \nabla \psi^*(\mathbf{r}) \right] - \frac{e^2}{m} |\psi|^2 \mathbf{A}(\mathbf{r}). \quad (6.16)$$

This is the second GL equation. These two GL equations in equation (6.162) and equation (6.164) are the main triumphs of GL theory. They enunciate the variation of the order parameter, $\psi(\mathbf{r})$, and the current density, $\mathbf{j}(\mathbf{r})$.

6.6.1 Coherence length and the penetration depth

Using the GL equations obtained above, we wish to derive expressions for some physical observables that are measured in experiments, such as, the coherence length and the penetration depth. Let us first talk about the coherence length.

Consider an inhomogeneous order parameter for a system generated by the presence of the boundary in the absence of an external magnetic field⁷. The superconducting (SC) region is denoted by $x > 0$. Thus, the order parameter is zero for the interface. For deriving expression for the coherence length, putting $\mathbf{A} = 0$ in the first GL equation (6.162),

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + \alpha |\psi| + \beta \psi^3 = 0. \quad (6.16)$$

It may be noted that α is negative in the superconducting state, which allows us to set, $\alpha = -|\alpha|$. Now defining the coherence length as,

$$\xi^2 = \frac{\hbar^2}{2m |\alpha|}. \quad (6.16)$$

6)

Also writing $(\beta / |\alpha|) \psi^2 = f^2$, we can write equation (6.165) as,

$$-\xi^2 f'' - f + f^3 = 0. \quad (6.167)$$

Multiply both sides by f' to get,

$$\begin{aligned} \frac{d}{dx} \left[-\frac{\xi^2 f'^2}{2} - \frac{1}{2} f^2 + \frac{1}{4} f^4 \right] &= 0 \\ \text{or, } -\xi^2 \frac{f'^2}{2} - \frac{1}{2} f^2 + \frac{1}{4} f^4 &= \text{constant}. \end{aligned} \quad (6.168)$$

Far from the boundary, f' (or ψ') should be zero, and $f^2 = 1$. Hence $|\psi|^2 = \frac{|\alpha|}{\beta}$.
Equation (6.168) now becomes,

$$\xi^2 (f')^2 = \frac{1}{2} (1 - f^2)^2. \quad (6.169)$$

The solution of the above equation is given by,

$$f = \tanh \left(\frac{x}{\sqrt{2}} \right).$$

Hence,

$$\psi = \left(\frac{|\alpha|}{\beta} \right)^{1/2} \tanh \left(\frac{x}{\sqrt{2}} \right)$$

This yields that ξ is the measure of the distance over which the order parameter responds to a perturbation. Also, since $\alpha = a(T - T_c)$, the temperature dependence of the coherence length can obtained as,

$$\xi(T) = \frac{\hbar^2}{2maT_c} \left(1 - \frac{T}{T_c} \right)^{-1/2} = \xi_0 \left(1 - \frac{T}{T_c} \right)^{-1/2}. \quad (6.170)$$

Thus $\xi(T)$ diverges as $(T - T_c)^{-1/2}$ as $T \rightarrow T_c$ from below. Also the temperature independent length, ξ_0 is given by,

$$\xi_0 = \frac{\hbar^2}{2maT_c}.$$

A little manipulation of the above formula yields,

$$\xi_0^2 = \frac{\hbar^2}{2m\epsilon_F} \left(\frac{\epsilon_F}{k_B T_c} \right)^2.$$

Thus ξ_0 is larger than $\left(\frac{\hbar^2}{2m\epsilon_F}\right)^{1/2}$, which is of the order of interparticle spacing, by a factor of $(\epsilon_F/k_B T_c)^2$. Remember, ϵ_F is usually of the order of a few eV (typically, 5–6 eV), and $k_B T_c$ for superconductors is of the order of a few meV. Thus ξ_0 is usually of the order of a few thousand lattice spacings.

Next, in order to arrive at an expression for the penetration depth, we examine the current expression from the second GL equation, that is equation (6.164). If we neglect the first term in comparison to the second term, then one obtains the London equation (discussed earlier), namely,

$$\mathbf{j}(\mathbf{r}) = \mu_0 \frac{1}{\lambda_L^2} \mathbf{A}(\mathbf{r}) \quad (6.171)$$

where,

$$\lambda_L^2 = \frac{mc^2}{4\pi e^2 |\psi|^2}$$

Again the temperature dependence of the penetration depth is obtained as,

$$\lambda_L = \left(\frac{mc^2 \beta}{4\pi e^2 a T_c} \right) \left(1 - \frac{T}{T_c} \right)^{-1/2} \quad (6.172)$$

λ_L has a similar divergence to that of $\xi(T)$. Also, a ratio of the two can be found as,

$$\kappa = \lambda_L / \xi = \frac{mc}{e\hbar} \left(\frac{\beta}{2\pi} \right)^{1/2} \quad (6.173)$$

As discussed earlier, the value of κ demarcates between various types of superconductors, namely, the type-I and type-II superconductors, which have very different magnetic properties.

6.7 Experimental determination of energy gap

The existence of a forbidden energy regime in superconductors has given us a simple way to understand the fact that below a certain excitation energy the bound state of the electrons cannot be broken. Thus the current in the superconducting state is mediated via the cooper pairs (also referred to as the *superelectrons*), and the normal current carried by the electrons is zero. More precisely, there are no unpaired electrons in the superconducting state. Here we describe a few of the early experimental methods below to determine the energy gap.

6.7.1 Absorption of electromagnetic radiation

This technique of exploring the energy gap was reported in 1957 by Glover and Tinkham [15]. They observed infrared transmission in thin superconducting films. It was proposed earlier that the electromagnetic (EM) waves of appropriate frequencies can break the ordered state of a

superconductor. Assuming a cooper pair has a binding energy given by, $E_B \sim 10^{-3}$ eV, one needs a radiation frequency $\nu \sim E_B/h = 2.4 \times 10^{11}$ Hz or 240 GHz. The wavelength of the waves is about 1 mm, for which neither generation of such frequencies, or detection methods were available in 1930s. Only two decades later, such EM waves could be produced.

In the experimental method, EM radiation is guided through a small cavity made in a superconducting material. Within the cavity, the radiation is reflected several times before it emerges through the cavity, and hence is detected. The stronger the absorption of the radiation from the walls of the cavity, the lower will be the output power. At a fixed temperature, $T < T_c$, that is, when the material is in the superconducting state, let the power in the superconducting and the normal states be I_s and I_n , respectively. $\frac{I_s - I_n}{I_n}$ is plotted as a function of frequency of the EM waves in figure 6.20. At small frequencies we see a distinct difference between detected power in the superconducting and the normal states, with the absorption being larger for the superconducting state. At certain threshold frequencies, the difference drops abruptly to zero. At even larger frequencies, the relative difference in the detected power stays at values that are vanishingly small. The abrupt drop sets in as soon as the quantum energy of the radiation is large enough to break the cooper pairs. Thus, there is an absorption edge at point *C* in figure 6.20. Similar absorption edges are also seen for semiconductors, however, the scale of the energy gap is much higher there, e.g. ~ 0.8 eV in Ge. As an example, for $h\nu > 2\Delta_0$, the energy gap practically vanishes and so does the absorption spectrum. This aids us in obtaining $2\Delta_0$, which is proportional to T_c . Some typical values for the superconductors, and their transition temperatures are given in table 6.1.

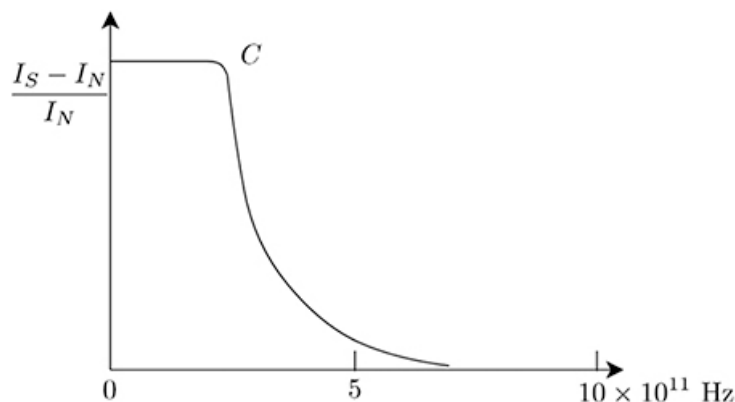


Figure 6.20. The absorption of electromagnetic radiation is shown as a function of frequency. The frequency at which the intensity almost vanishes yields the superconducting gap 2Δ .

Table 6.1. 2Δ and T_c are reported for a few conventional superconductors. The gap is of the order of a few meV.

	T_c	$2\Delta(\text{meV})$
Nb ₃ Sn	18	6.55
MgB ₂	40	10
Rb ₃ C ₆₀	30	12
ErRh ₄ B ₄	9	3

6.7.2 Ultrasound absorption

Sound waves denote propagation of phonons in a material. Most measurements with sound waves are carried out with typical frequencies of the order of 10 MHz. These frequencies correspond to energies that are much smaller than the superconducting gap. Only very close to T_c , where the energy gap Δ approaches zero, do sound energies at the MHz frequency range become comparable to the gap. Unpaired electrons below T_c interact with the lattice excitations leading to damping. Below T_c , as the temperature is decreased, the number of unpaired electrons decreases rapidly, thereby causing reduced damping of the sound intensity. A schematic plot of the ultrasound absorption spectra (in arbitrary units) is presented as a function of the reduced temperature, T/T_c . As T tends to T_c from below, the Cooper pairs start dissociating yielding an increasingly large number of unpaired electrons in the system. These electrons absorb the sound waves leading to an enhancement in the absorption intensity (see figure 6.21). Since at a given temperature, the number of unpaired electrons depends on the energy gap, the latter can be determined from such absorption experiments.

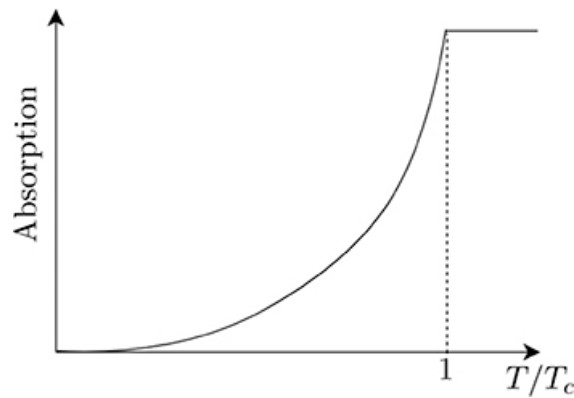
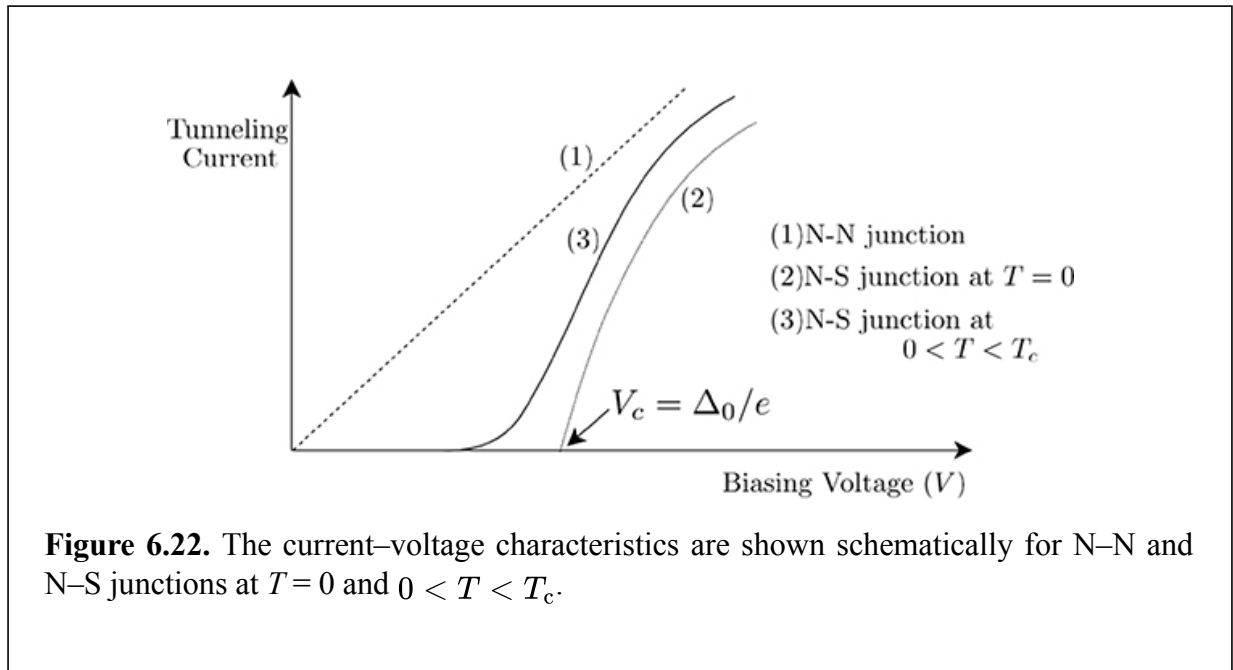


Figure 6.21. Schematic depiction of the ultrasound absorption intensity is presented as a function of T/T_c . The absorption is low in the superconducting phase owing to the absence of unpaired electrons.

6.7.3 Tunneling experiment

In 1961, I Giaever [16] pointed out the possibility of determining the energy gap by means of tunnelling experiments. The experiments involve observation of tunnelling current across a metal-insulator-semiconductor (MIS) junction. The metal and the insulating materials are usually taken as Al and Al_2O_3 , respectively. Al_2O_3 is known to be a good insulator which can be fabricated nearly perfectly with a thickness of only a few nanometers.

One can understand the tunnelling effect without a detailed calculation. Consider a particle that incidents on a barrier with an energy lower than the barrier height. While a classical particle is unable to travel to the other side of the barrier, it is possible for a quantum mechanical particle (because of its wave nature) to do so. Considering the wave character of the particle, when a wave is incident on the surface separating two media into which it cannot enter, the wave must be totally reflected. We understand intuitively that being a wave, it can penetrate up to a small distance into the forbidden region with its amplitude decreasing exponentially. The decrease is faster and complete within the barrier if the width and height of the barrier are large. However, for a sufficiently thin barrier there is a finite possibility that the wave can penetrate into the barrier (of the order of de Broglie wavelength), and propagate into the medium (lead) on the other side. If a superconductor is introduced in one or both sides of the barrier, the current as a function of the biasing voltage (I - V) characteristic is bound to be different. This happens because an energy gap appears in the superconducting state. So no tunnelling current can flow below a certain critical voltage, given by, $V_c = \Delta/e$. From figure 6.22 it is clear that from V_c , the value of Δ_0 can be determined.



However, at the normal-superconductor junction there is an additional phenomenon which contributes to the tunnelling current. Here especially we are interested in a situation in which the biasing voltage is much smaller compared to the energy gap, Δ , in a superconductor. Thus,

in usual circumstances, no tunnelling can occur. Thus, for an electron arriving at the interface of a normal-superconductor junction, two possible processes can take place (see figure 6.23), namely:

- (i) a normal reflection in which an electron is simply reflected back giving rise to no net current;
- (ii) Andreev reflection which is an anomalous reflection of an electron. What actually happens here is the following. An electron comes at the interface, instead of undergoing a reflection
 - (a) a Cooper pair is added in the superconducting region, it has an energy Δ ;
 - (b) the incident electron is annihilated;
 - (c) a hole is reflected back along the original path of the electron.

This is known as Andreev reflection. The incident, and the reflected quasiparticles have approximately equal wave vectors, which are given by,

$$\begin{aligned} k^{\text{el}} &= k_F + \frac{\epsilon}{\hbar v_F} \\ k^n &= k_F - \frac{\epsilon}{\hbar v_F} \end{aligned} \quad (6.17)$$

but have opposite directions of motion as appears from the sign of the group velocity $\frac{1}{\hbar} \frac{\partial \epsilon}{\partial k}$. The momentum is conserved up to the order,

$$\hbar |k^{\text{el}} - k^n| \leq \hbar/\xi$$

where $\xi = \frac{\hbar v_F}{\pi \Delta_0}$, which is the coherence length.

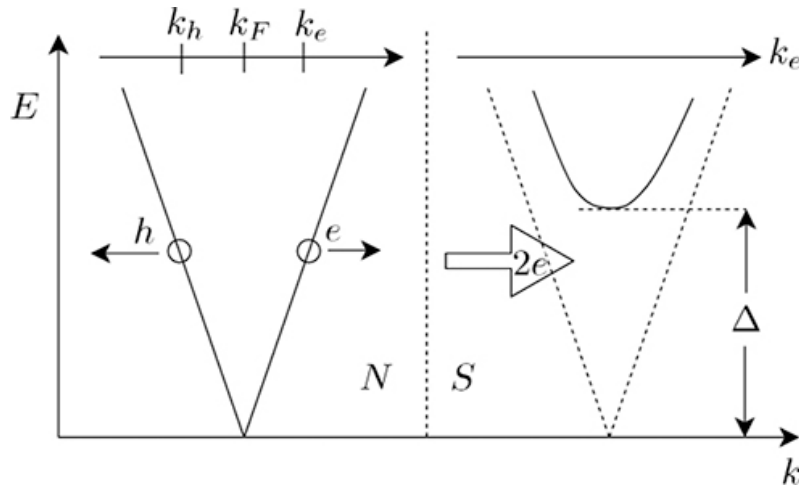


Figure 6.23. A schematic diagram of the dispersion E versus k corresponding to the Andreev reflection at the normal-superconductor (N-S) junction is shown. A Cooper pair

is created at the boundary which can propagate at low bias inside a superconductor. The electron is specularly reflected as a hole on the metallic side.

The energy is also conserved in the process. The minimum energy of a Cooper pair has a value $2\epsilon_F$, the incident electron has energy $\epsilon_F + \epsilon$, and that of the reflected hole is $\epsilon_F - \epsilon$. In the process, a charge $2e$ is transferred from the normal to the superconducting side of the barrier, which is equivalent to a Cooper pair being injected into the superconductor.

We can understand the tunnelling problem in the following manner. The number of electrons tunnelling per unit time with an energy E from the left (say, a metal) to the right (a superconductor) is proportional to the number of occupied states on the left, namely, $N_1(E)f(E)$, and that to the unoccupied states on the right, $N_2(E')(1 - f(E'))$. $N_1(E)$ and $N_2(E')$ denote DOSs for the metallic and the superconducting regions, respectively. In the presence of the biasing voltage, V , the energies are given by,

$$E' = E + eV$$

Thus the unoccupied states on the right are denoted by,

$$N_2(E + eV)(1 - f(E + eV)) \quad (6.17)$$

The tunnelling current is proportional to both these independent process, which can be written as, 5)

$$I_{1 \rightarrow 2} \propto \int_{-\infty}^{\infty} N_1(\epsilon)f(\epsilon)N_2(\epsilon + eV)(1 - f(\epsilon + eV)) d\epsilon \quad (6.17)$$

This tunnelling current is a measurable quantity in experiments. 6)

For a normal metal–superconductor junction, the current, $I_{N \rightarrow S}$ can be obtained by replacing $N_1 \rightarrow N_N$ and $N_2 \rightarrow N_S$. A schematic plot of the tunnelling current for the junctions, namely, metal–metal (N – N) and metal–superconductor (N – S) both at $T = 0$ and for $T < T_c$ as a function of the bias voltage are presented in figure 6.22. The threshold potential yields the energy gap. A differential conductance, $\frac{dI}{dV}$, which is proportional to the density of states at the Fermi level is often obtained in experiments.

6.7.4 Unconventional superconductivity

There are several superconductors, that have been discovered over the last few decades, which challenge the well-established paradigm of conventional superconductivity. Quite a few of them, including the high- T_c cuprates, pose anomalies owing to features that are unfamiliar in the context of Fermi liquid theory. As a first signature, Uemura presented the plot of T_c versus T_F (both in log scale) that yields a straight line, while the conventional superconductors are far away from this line. This is called the Uemura plot and its freehand sketch is presented in figure 6.24.

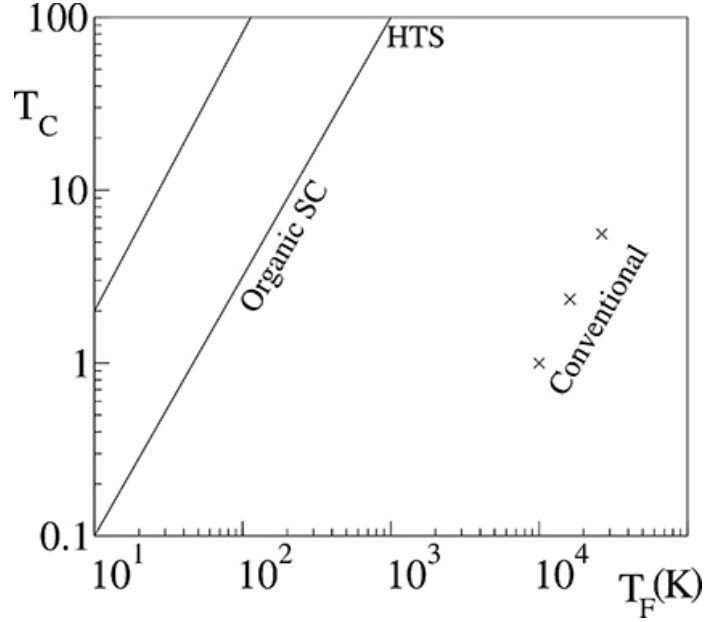


Figure 6.24. Uemura plot showing the superconducting transition temperature, T_c versus the Fermi temperature, T_F (both in log scale). The positions of different superconductors are shown. The conventional (BCS) superconductors lie far away from the unconventional superconductors.

6.7.5 High- T_c cuprates

In 1986, 75 years after the discovery of superconductivity, G Bednorz and K Muller at IBM, Zurich demonstrated superconductivity in a perovskite structured lanthanum (La) based copper oxide material which registered a T_c of about 39 K for which the Nobel prize was awarded in 1987. It was a remarkable discovery, as it later allowed chemical substitution in perovskite cuprates to push the T_c well beyond the liquid nitrogen temperature (77 K), which is a much cheaper and easily accessible quantity than liquid helium. Later in 1987, La was substituted with yttrium (Y) in the form of $\text{YBa}_{2-x}\text{Cu}_3\text{O}_{7-x}$ which showed a T_c of about 92 K. The materials show highest T_c when they are slightly oxygen deficient, that is, when $x \simeq 0.15$. Superconductivity disappears at larger values of x , which is also accompanied by a structural phase transition when YBCO changes from an orthorhombic structure to a tetragonal structure. Subsequently thallium (Th) and mercury (Hg) based superconductors show even higher transition temperatures. These superconductors are generally type-II superconductors which show gradual change in temperature as a function of the external magnetic field.

A large volume of research has taken place since its discovery [17], however, owing to its ill understood normal state, a comprehensive understanding of the microscopic phenomena remained elusive. The electronic pairing mechanism is itself questionable and possibly other pairing symmetries, other than the conventional s -wave discussed in the context of BCS theory

play crucial roles. Even after several decades of active research, several anomalies that questioned the established paradigm of superconductivity, were left without clarification. Similar unanswered questions exist for superconductivity found in heavy fermion compounds, iron based metals and more recently in twisted bilayer graphene.

The cuprate superconductors have a generic phase diagram (see figure 6.25) which itself poses a bunch of surprises. The proximity of the Mott insulating state and the superconducting phase, a pseudogap phase in the underdoped phase are among the few, while the overdoped regime is probably a familiar metal. We shall not prolong the discussion here, and rather motivate readers to look at the review mentioned in the footnote and the references therein. A few of the CuO₂ superconductors are listed here along with transition temperatures (table 6.2).

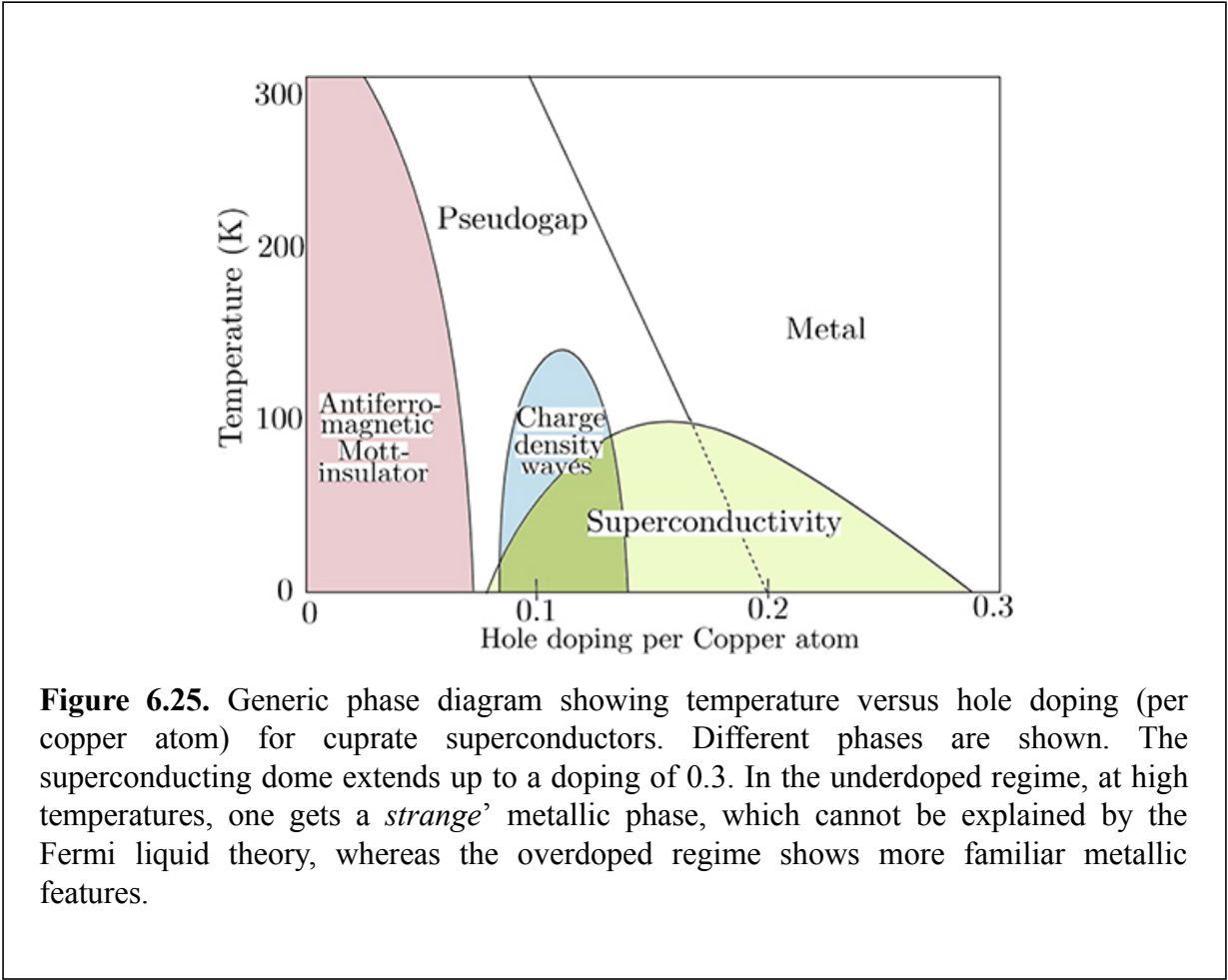


Figure 6.25. Generic phase diagram showing temperature versus hole doping (per copper atom) for cuprate superconductors. Different phases are shown. The superconducting dome extends up to a doping of 0.3. In the underdoped regime, at high temperatures, one gets a *strange*’ metallic phase, which cannot be explained by the Fermi liquid theory, whereas the overdoped regime shows more familiar metallic features.

Table 6.2. T_{cS} of some typical cuprate superconductors are shown.

	T_c
$\text{La}_{2-x}\text{Ba}_x\text{CuO}_4$	30 K

	T_c
$\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$	39 K
$\text{YBa}_2\text{Cu}_3\text{O}_7$	92 K
$\text{Bi}_2\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	110 K
$\text{Te}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	125 K
$\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_8$	134 K

The compounds that are of special interest are $\text{Bi}_2\text{CaSr}_2\text{Cu}_2\text{O}_8$ (called Bi – 2122 compounds) and $\text{Tl}_2\text{CaBa}_2\text{Cu}_2\text{O}_8$ (called Tl – 2122 compounds). Because of their large T_c (>100 K), they have different numbers of cuprate layers and are denoted by a general formula $\text{A}_2\text{Ca}_n\text{Y}_2\text{Cu}_n\text{O}_{2n+4}$ with $\text{A} = \text{Bi}, \text{Tl}$ and $\text{Y} = \text{Sr}, \text{Ba}$, etc. The ones with Bi and Sr are more widely studied and are described by the formula $\text{Bi}_2\text{Sr}_2\text{Ca}_{n-1}\text{Cu}_n\text{O}_{2n+4+\delta}$ where $n = 1 \rightarrow 3$ and δ denotes the oxygen doping. These are called BSCCO (pronounced as BISCO). $n = 1, n = 2$ and $n = 3$ have a T_c of 33 K, 96 K, and 108 K, respectively. Similarly, for the Tl family (denoted by $\text{TlBa}_2\text{Ca}_{n-1}\text{Cu}_n\text{O}_{2n+3}$) $n = 1, 2$ and 3 correspond to $T_c \sim 14$ K, ~ 80 K and ~ 120 K, respectively.

There are numerous examples of anomalies noted in various experiments on the cuprate superconductors. For example, consider ultrasound attenuation measurements. In conventional (BCS) superconductors, the ultrasound attenuates exponentially due to loss of unpaired electrons (in favour of forming Cooper pairs). The coefficient of attenuation α is proportional to $e^{-\Delta/K_B T}$ where Δ denotes the superconducting energy gap. The scenario is not too different in the cuprate family. However, the ultrasonic attenuation below T_c is found to be lower than that of the BCS superconductors, possibly owing to the formation of pseudogap. However, the longitudinal sound velocity and the elastic moduli show discontinuities in most of the cuprate superconductors (except for the Tl-based family). The discontinuity points towards a structural change and ‘*lattice softening/hardening*’ at the superconducting transition. The anomaly in the elastic moduli points toward a collective ordering of the oxygen atoms in the CuO_2 layers. There are many such examples which show significant digression compared to the behaviour observed for conventional superconductors. Apart from being beyond the scope of the present discussion, there are many good reviews on the subject, and interested readers are encouraged to look at some of these. In the following we review the most significant feature of the cuprates which eluded a microscopic theory so far, namely the formation of a pseudogap phase.

6.8 The pseudogap phase

The nature of the spectral gap in superconductors and how it vanishes (or changes) across a transition can be detected by angle resolved photoemission spectroscopy (ARPES) experiments. In the context of high T_c superconductors, high energy photons from a synchrotron source with an energy ≈ 20 eV made to incident on the surface of the single crystal. As a result, a photoelectron is ejected at an angle θ normal to the surface of the crystal. The kinetic energy of the ejected electron is measured via an electron spectrometer whose energy resolution is of the order of a few meV. The ARPES data for high T_c cuprates are thoroughly analysed and reviewed in [18, 19].

The parallel component of the momentum ($\mathbf{k}$) is balanced by the momentum of the excitations created inside the sample. By varying the detecting angle θ between the normal and the position of the detector it is possible to map out the momentum dependence of the filled states of the crystal. Thus the intensity of the photoelectron distribution will demonstrate a sharp peak corresponding to the parallel component of the momentum with an energy ω which is given by,

$$I(\mathbf{k}, \omega) = I_0(\mathbf{k})f(\omega)A(\mathbf{k}, \omega) \quad (6.17)$$

where the energy is measured with respect to the Fermi level [19] and $A(\mathbf{k}, \omega)$ denotes ⁷⁾ the spectral function of the excitations that corresponds to the hole that is left behind. $I_0(\mathbf{k})$ is the arbitrary intensity factor that preserves the dimensionality of the equation and $f(\omega)$ is the Fermi distribution function, $f(\omega) = \frac{1}{e^{\beta(\omega-\mu)}+1}$, $\hbar = 1$ and β is the inverse temperature.

The important quantity in the above equation (6.177) is the spectral function which contains the information about the electron–electron interaction present in the material via the self energy $\Sigma(\mathbf{k}, \omega)$, which is in general a complex quantity. The spectral function $A(\mathbf{k}, \omega)$ is expressed in terms $\Sigma(\mathbf{k}, \omega)$ as [20],

$$A(\mathbf{k}, \omega) = \frac{\Sigma''(\mathbf{k}, \omega)}{(\omega - \epsilon_k - \Sigma'(\mathbf{k}, \omega))^2 + \Sigma''(\mathbf{k}, \omega)} \quad (6.17)$$

Where Σ' and Σ'' represent the real and the imaginary parts of the self energy and ϵ_k ⁸⁾ denotes the single-particle energies.

The ARPES spectra measure $A(\mathbf{k}, \omega)$ and show a broad background as a function of ω that extends to lower energies which yields the energy loss of the photoelectrons within the crystal. For a recent review see reference [21]. However, as a result of strong interparticle interactions, which dominate in these cuprate superconductors, this broad background is attributed due to the electronic correlations. The energy of the emergent photoelectrons yields a sharp coherent peak at the single-particle energies given by ϵ_k and broadens rapidly for state below the Fermi level as the detection angle θ is varied. As said earlier, the loss of coherence is due to the presence of strong electronic correlations [21] and hints towards the breakdown of the Fermi liquid picture (see section 1.5). The optical conductivity data support the loss of coherence as well. The sharp peak at the Fermi level as a function of the detection angle θ maps out the Fermi surface as it yields the information on the momentum (and hence the energy) of the emitted photoelectron. The same experiment can be repeated at various doping levels. Mostly

in cuprates, at optimal doping and at the overdoped regime, the presence of the sharp peak in the ARPES spectrum confirms a Fermi liquid picture to be valid. However, a noticeable broadening is noted in the underdoped regime at the Fermi level and that implies breakdown of Fermi liquid theory (FLT).

Both at $T < T_c$ (superconducting state) and $T > T_c$ (normal state), there are deviations from the FLT which are attributed to the formation of pseudogap in cuprates. The pseudogap is most likely related to and has the same symmetry as the superconducting gap (believed to be d -wave like). However, unlike a conventional superconducting gap, the magnitude of the pseudogap is temperature independent. ARPES and other experiments indicate that the magnitude of the gap does not change significantly as the temperature is lowered through T_c . Thus, in the case of the cuprates, unlike a conventional superconducting gap, the gap starts as a pseudogap in the normal state and evolves into the superconducting gap below T_c . In other words, $\Delta(T)$ does not go to zero at $T = T_c$.

The pseudogap appears in the underdoped regime of the phase diagram and weakens towards optimal doping. A weak pseudogap is still present at optimal doping⁸, but rapidly disappears thereafter. The spectral gap is purely superconducting in the overdoped region. This is illustrated in the phase diagram shown in figure 6.25 where T^* approaches T_c just into the overdoped region. T^* is temperature dependent in the underdoped regime. Well into the overdoped region, the pseudogap merges with the superconducting gap, which means that T^* merges with T_c , and there remains no trace of the pseudogap in the normal state.

To wind up the discussion, we note down a few salient features. They are:

- (i) the pseudogap is different than the superconducting gap in the underdoped phase of cuprates, and gradually evolves into the superconducting gap at larger doping levels;
- (ii) both the pseudogap and superconducting gaps are believed to have d -wave symmetry;
- (iii) the crossover temperature T^* (shown in figure 6.25) merges with T_c in the overdoped region of the phase diagram.

6.8.1 Summary and outlook

Let us include a brief recap of the topics discussed. By and large we have confined ourselves to the study of conventional superconductors. We have highlighted several distinguishing properties of superconductors, such as Meissner effect, isotope effect, distinction between perfect conductors and superconductors, flux quantization, penetration depth, and the coherence length, type-I and type-II superconductors, etc. Hence we have discussed magnetic and thermodynamic properties of superconductors, where the former shows phase diagram as a function of the external magnetic field that encodes a phase transition, either directly from a superconductor to a metal, or the same intervened by a mixed phase that admits the magnetic flux lines (in unit of a constant flux quantum) to penetrate the sample. Most of the well-known conventional superconductors are of the latter variety. The thermodynamic properties, such as the specific heat, show a jump of fixed magnitude at the transition point and are a generic feature of a superconducting phase transition.

Hence we have embarked on the famous BCS theory, where we have discussed Cooper's instability that leads to the formation of bound pairs of electrons mediated via phonons and is known to provide the microscopic origin of the pairing phenomena. This was followed by a

detailed description which yields an appropriate estimation of the temperature dependence of the spectral gap, and hence the transition temperature, where the latter provides enumeration of different superconductors. Among the crucial properties, we have outlined the calculation of the specific heat, computed the behaviour of the specific heat and the jump therein at the transition temperature, and discussed Meissner effect.

We have also included a brief recap of the Ginzburg–Landau (GL) theory which is a phenomenological theory of superconductivity proposed by Ginzburg and Landau in the early 1940s, and is broadly applicable to all second-order phase transitions. Apart from discussion of the two GL equations, we have estimation of two important length scales that characterize a superconductor, namely the coherence length and the penetration depth. By no means is the above discussion complete, and readers are encouraged to look up more detailed notes on the subject.

We have also described in brief a few experimental methods of estimating the energy gap in superconductors. These are electromagnetic, ultrasound absorption experiments, and measuring the tunnelling spectra in junction systems, involving a normal metal and a superconductor (N–S junction), or a metal between two superconductors (SNS junction), etc.

To wind up, we have mentioned very briefly about the unconventional superconductors that have been discovered over the last two and a half decades. There are a number of issues that necessitates going beyond the BCS paradigm. Once again readers are encouraged to look at literature, including some excellent review articles on the subject.

References

- [1] Kamerlingh Onnes H *Research Notebooks* **56** 57 Kamerlingh Onnes H *Archive* (Leiden, the Netherlands: Boerhaave Museum); Kamerlingh Onnes H 1911 *Commun. Phys. Lab. Univ. Leiden* 122 124; Kamerlingh Onnes H 1913 *Commun. Leiden* I33atoI33d
- [2] Hulm J K and Matthias B T 1951 *Phys. Rev.* **82** 273
Hulm J K and Matthias B T 1953 *Phys. Rev.* **89** 439
Hulm J K and Matthias B T 1952 *Phys. Rev.* **87** 799
Hulm J K and Matthias B T 1953 *Phys. Rev.* **92** 874
- [3] Becker R, Heller G and Sauter F 1933 Über die Stromverteilung in einer supraleitenden Kugel *Z. Phys.* **85** 772–87
- [4] London F and London H 1935 *Proc. R. Soc. A* **149** 71
- [5] Deaver B S and Fairbank W M 1961 *Phys. Rev. Lett.* **7** 43
- [6] Tinkham M 1973 *Introduction to Superconductivity* 2nd edn (Dover: New York)
- [7] Ramakrishnan T V and Rao C N R 1999 *Superconductivity Today* (Hyderabad: Universities Press)
- [8] Kittel C 2004 *Introduction to Solid State Physics* 8th edn (Hoboken, NJ: Wiley)
- [9] Fröhlich H 1950 *Phys. Rev.* **79** 845
- [10] Cooper L N 1956 *Phys. Rev.* **104** 1189
- [11] Sakurai J J and Napolitano J 2011 *Modern Quantum Mechanics* 2nd edn (Reading, MA: Addison-Wesley)
- [12] Reif F 1965 *Fundamentals of Statistical and Thermal Physics* (New York: McGraw-Hill)
- [13] Ginzburg V L and Landau L D 1950 *Zh. Eksp. Teor. Fiz.* **20** 1064 English translation in: Landau L D 1965 *Collected Papers* (Oxford: Pergamon Press) p 546
- [14] Fetter A L and Hohenberg P C 1969 *Superconductivity* 1st edn ed R D Parks (Boca Raton, FL: CRC Press)
- [15] Tinkham M and Glover R E 1958 *Phys. Rev.* **110** 778
- [16] See the Nobel lecture by I Giaever available at: <http://www.nobelprize.org/prizes/physics/1973/giaever/lecture/>
- [17] Verma C M 2020 *Rev. Mod. Phys.* **92** 031001
- [18] Shen Z-X *et al* 1993 *Phys. Rev. Lett.* **70** 3999
- [19] Randeria M and Campuzano J C *Varenna Lectures* (arXiv:9709107)
- [20] Fetter A L and Walecka J D 1971 *Quantum Theory of Many Particle Systems* (New York: McGraw-Hill)
- [21] Timusk T and Statt B 1999 *Rep. Prog. Phys.* **62** 61

¹The normal state refers to the metallic state which is expected as superconductivity is lost.

²The working principle of STM is based on the measurement of electronic tunnelling current that depends exponentially on the distance between the tip and the sample. The position of a tip on top of a flat surface of the sample can be controlled with subatomic resolution.

³In order for the quasiparticle operators (γ) to diagonalize, the idea is to make coefficients of $\gamma\gamma$ or $\gamma^\dagger\gamma^\dagger$ equal to zero and retain $\gamma^\dagger\gamma$. This happens when $2\xi_k u_k v_k + \Delta_k^* v_k^2 - \Delta_k u_k^2 = 0$.

⁴This is the numerical solution of the gap function, $\Delta(T)$ shown in figure 6.18.

⁵For the dc conductivity, we can take $\omega \rightarrow 0$.

⁶It may be mentioned here that we have introduced the Ginzburg–Landau theory very briefly. In no way should it be considered as complete. For a detailed discussion, see *Theory of type-II superconductors* in reference [14].

⁷To derive the coherence length, the presence of a magnetic field is not essential.

⁸Optimal doping corresponds to highest T_c .